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RESEARCH ARTICLE

A Generalized Spatio-Temporal Causal Knowledge Graph Framework With Large Language Model-Enhanced Reasoning for Urban Event Management

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ABSTRACT Urban event management faces critical challenges in processing unstructured citizen complaints because of the limitations of traditional extraction models in handling colloquial text, static nature of conventional knowledge graphs that overlook spatiotemporal risk propagation, and brittleness of handcrafted symbolic rules that fail to generalize to novel scenarios. To address these issues, this study presents ST-CKG, a generalized spatiotemporal causal knowledge graph framework with Large Language Model (LLM)-enhanced reasoning. The framework integrates three tightly coupled components: a hybrid information extraction module that combines fine-tuned models with an LLM-based verifier to produce confidence-calibrated structured knowledge; a spatiotemporal causal hypergraph construction module that encodes spatial proximity and temporal causal hyperedges to model cascading urban risks; and a neuro-symbolic reasoning engine that distills expert-defined risk causal chain rules into a graph attention network, enabling interpretable and generalizable inference over incomplete or unseen event configurations. Extensive experiments conducted on over 2,000 real-world citizen complaints demonstrated that the hybrid extraction module outperformed standalone baselines by 3–5 percentage points in the F1-score, achieving 91% urgency assessment accuracy, significantly surpassing pure rule-based and pure graph neural network variants in rule-uncovered scenarios. Ablation studies have confirmed the indispensability of each component. The ST-CKG framework offers a replicable paradigm for bridging unstructured civic narratives with structured actionable governance decisions in smart cities.

INDEX TERMS Causal reasoning, graph neural networks, knowledge graphs, large language models, neuro-symbolic reasoning, smart cities, spatio-temporal analysis, urban event management.

I. INTRODUCTION

Rapid urbanization of the 21st century has fundamentally transformed municipal governance paradigms worldwide. As cities expanded in terms of both population density and functional complexity, the volume of citizen complaints, service requests, and incident reports received through

government hotlines surged to unprecedented levels [1]. In megacities such as Beijing, the 12345 citizen service hotline processes millions of appeals annually across categories, including municipal infrastructure failures, public safety hazards, environmental violations, and housing disputes, each requiring timely assessment, accurate routing, and effective cross-departmental coordination [2]. The transformation of unstructured civic narratives into structured, actionable intelligence represents a critical challenge for smart-city initiatives.

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Knowledge Graphs (KGs) have emerged as a powerful paradigm for structuring heterogeneous urban information, enabling semantic querying and automated reasoning across organizational silos [3]. Recent advances have extended static KGs with spatiotemporal semantics to support context-rich reasoning over location-grounded and time-sensitive facts [4], [5]. In the urban computing domain, Spatio-Temporal Graph Neural Networks (STGNNs) have demonstrated strong predictive learning capabilities for traffic forecasting, public safety monitoring, and environmental sensing [6]. However, existing urban Knowledge Graph (KG) approaches predominantly focus on physical infrastructure and mobility patterns, paying insufficient attention to the causal chains of sociotechnical risks embedded in citizen complaints and narratives that encode the most immediate governance demands [7]. Furthermore, conventional KGs remain largely static, requiring manual curation for updates and a lack of capacity for real-time adaptation to dynamically evolving urban events [8].

Parallel to these developments, large language models have demonstrated remarkable capabilities in natural language understanding, prompting researchers to explore their potential in KG construction and urban analytics [9], [10]. Recent studies have integrated LLM-assisted entity extraction with temporal KGs incorporating geocoded toponyms, achieving competitive performance on long-tailed corpora [11]. Similarly, multimodal urban space frameworks have leveraged Large Language Models (LLMs) to identify relationships between incidents across heterogeneous data sources [12]. However, a critical limitation of LLMs—their tendency to generate hallucinated content—poses severe risks in governmental environments, where factually incorrect outputs can lead to policy errors and a breach of public trust [13], [14]. Comprehensive evaluations have confirmed that hallucinations persist across diverse named entity recognition tasks and domains, although their severity can be mitigated through careful output verification mechanisms [15]. These findings motivate a hybrid approach that combines the precision of fine-tuned models with the semantic coverage of LLMs, while maintaining mechanisms for hallucination detection.

The emerging field of neurosymbolic AI offers a promising direction for bridging the gap between data-driven learning and expert-defined symbolic rules [16], [17]. Recent surveys have comprehensively reviewed neural-symbolic reasoning methods over KGs, highlighting how the integration of symbolic rules with neural networks can overcome the limitations of purely data-driven or purely rule-based approaches, particularly when dealing with incomplete and noisy data [18], [19]. In particular, Graph Attention Networks (GATs) have demonstrated strong capabilities for KG reasoning tasks, including link prediction and entity classification through adaptive attention mechanisms [20], [21]. Nevertheless, the application of neurosymbolic reasoning in urban event management, where expert-defined rules governing risk causality

and departmental jurisdiction coexist with noisy colloquial complaint narratives, remains largely unexplored.

To address the aforementioned limitations, this study proposes a generalized spatiotemporal causal knowledge graph framework with Large Language Model (LLM)-enhanced reasoning for intelligent urban event management. The proposed framework integrates hybrid event extraction, spatio-temporal hypergraph modeling, neurosymbolic reasoning, and confidence-aware knowledge fusion to support urgency prediction and interdepartmental collaboration recommendations in complex governance environments.

The main contributions of this work are summarized as follows:

- 1) A hybrid LLM-enhanced event extraction framework was developed to address semantic ambiguity and information incompleteness commonly observed in heterogeneous municipal complaint data. This framework integrates fine-tuned domain-specific extraction models, LLM-driven semantic verification, and confidence-aware fusion mechanisms. This design improves the robustness of structured knowledge acquisition, while mitigating hallucination risks in urban governance scenarios.

- 2) A generalized spatiotemporal causal hypergraph representation is constructed to model complex urban governance relationships involving spatial regions, temporal evolution, event categories, and interdepartmental collaboration patterns. Compared to conventional pairwise graph representations, the proposed hypergraph structure captures high-order relational dependencies and causal propagation behaviors more effectively.

- 3) A neurosymbolic reasoning framework combining Semantic Web Rule Language (SWRL)-based symbolic inference and Hypergraph Attention Network (HyperGAT)-based relational learning is proposed to enhance both interpretability and generalization capability. Through adaptive rule distillation and neural relational reasoning, the framework enables effective prediction of both rule-covered and rule-uncovered governance events, addressing a key limitation of purely symbolic approaches.

- 4) Extensive experiments on real-world municipal governance datasets, including ablation analysis, sensitivity evaluation, and cross-theme generalization experiments, demonstrated that the proposed framework achieves robust generalization capability and stable reasoning performance under dynamically evolving governance environments.

The remainder of this paper is organized as follows. Section II reviews related work on knowledge graph construction, spatiotemporal reasoning, and neuro-symbolic AI. Section III details the proposed ST-CKG framework, covering the hybrid extraction module, spatio-temporal hypergraph construction, and neurosymbolic reasoning engine. Section IV presents the experimental setup and comprehensive evaluation results. Section V discusses the implications, limitations, and directions for future research. Finally, Section VI concludes the paper.

II. RELATED WORK

This section reviews prior research across four thematic areas that collectively inform the design of the proposed ST-CKG framework: (A) knowledge graph construction and its applications in smart cities, (B) spatio-temporal knowledge representation and reasoning, (C) large language models in information extraction and knowledge engineering, and (D) neuro-symbolic reasoning over knowledge graphs. For each area, we identified critical gaps that motivated this study.

A. KNOWLEDGE GRAPH CONSTRUCTION AND SMART CITY APPLICATIONS

Knowledge graphs have emerged as a foundational technology for structuring heterogeneous data into interconnected entity-relation triples, facilitating semantic querying, knowledge sharing, and automated reasoning across diverse application domains [3]. Construction pipelines typically encompass ontology design, information extraction, entity resolution, and graph population, each of which presents distinct challenges when applied to domain-specific corpora [8]. In the smart city domain, KGs have been deployed across multiple verticals, including transportation management, urban planning, parking optimization, and environmental monitoring [7], [22]. A recent survey by Wang et al. [22] comprehensively reviewed KG-related research in the smart city domain from four perspectives: smart city ontology, urban data processing, and urban KG construction and their applications, highlighting the growing adoption of domain-specific KGs for urban intelligence. Pan et al. [32] provided a roadmap for unifying LLMs with KGs, identifying structured knowledge integration as a key enabler for intelligent urban systems.

Despite these advances, the application of KGs to citizen complaint management and community governance remains underexplored. Lei and Li [23] demonstrated the feasibility of using knowledge graphs and pretrained models for urban public crisis prediction by integrating heterogeneous datasets from social media, public records, and GIS, highlighting the potential of AI-driven approaches for risk assessment in urban governance. However, their approach focused on general crisis prediction rather than the specific task of risk causal chain extraction from citizen-appeal text. More broadly, the integration of KGs with real-time urban event streams, particularly those originating from government hotspots, has not been systematically addressed in literature [2].

B. SPATIO-TEMPORAL KNOWLEDGE REPRESENTATION AND REASONING

The incorporation of spatial and temporal semantics into knowledge graphs has attracted significant research attention, driven by the recognition that many real-world facts are inherently grounded in geographic locations and time intervals [4], [5]. Recent surveys have comprehensively reviewed spatiotemporal knowledge graph models covering both construction and embedding methods [30], [31].

Temporal Knowledge Graphs (TKGs) extend traditional KGs by associating facts with timestamps, thereby enabling reasoning over event sequences and temporal dependencies. The LMS model proposed by Zhang et al. [24] learns multigraph structures within TKGs to capture concurrent and evolutionary patterns along timestamps, thereby demonstrating superior performance on event-based benchmark datasets for extrapolative reasoning. Their work underscores the importance of modeling both spatial and temporal interactions to accurately forecast future events.

In the geographic domain, AugGKG, presented by Han et al. [25], introduced a grid-augmented geographic KG representation based on the GeoSOT global subdivision grid model and time-slice subgraph architecture, thereby enabling an efficient multi-objective spatiotemporal query answering. Lei et al. [4] proposed a hazard event ontology model with multi-scale spatiotemporal coding to represent urban storm surge hazard information, demonstrating the effective integration of heterogeneous hazard data across spatial and temporal scales. Ning et al. [2] constructed the UUKG dataset, a unified urban KG that connects administrative borders, points of interest, and road segments, specifically designed to support knowledge-enhanced spatiotemporal predictions. In the broader urban computing landscape, Jin et al. [6] provided a comprehensive survey of STGNN technologies for predictive learning, covering transportation, the environment, public safety, and healthcare applications. Recent advances in Temporal Knowledge Graph (TKG) reasoning have explored dynamic evolution and relation perception mechanisms [38], as well as complex causal relations [39], further enriching the spatio-temporal modeling toolbox. Han et al. [42] proposed a grid neighborhood-based graph convolutional network for spatio-temporal KG reasoning, demonstrating the effectiveness of grid-structured spatial encoding for geographic knowledge.

Nevertheless, the existing spatiotemporal KG approaches predominantly address physical infrastructure monitoring, traffic flow prediction, and environmental sensing. The modeling of sociotechnical risk propagation, in which citizen complaints regarding infrastructure failures, safety hazards, and regulatory violations exhibit complex spatial clustering and temporal recurrence, has received comparatively little attention. Critically, none of the surveyed methods incorporated risk causal chains as first-class modeling constructs within spatiotemporal graph representations.

C. LARGE LANGUAGE MODELS IN INFORMATION EXTRACTION AND KNOWLEDGE ENGINEERING

The advent of large language models has introduced transformative capabilities for knowledge graph construction, particularly in automating the extraction of entities, relations, and events from unstructured text [9], [10]. LLM-based approaches offer distinct advantages over traditional fine-tuned models: zero-shot and few-shot generalization across domains, ability to handle colloquial and noisy text, and rich

semantic understanding that captures implicit relations often missed by supervised extractors [9]. Lan et al. [40] proposed an LLM-based few-shot method for constructing academic knowledge graphs, achieving competitive performance with minimal labeled data, further demonstrating the feasibility of LLM-driven KG construction in specialized domains. Recent work has demonstrated the feasibility of LLM-driven KG construction across diverse fields, including the scientific literature [27] and semantic communication systems [26]. The SF-GPT framework proposed by Sun et al. [27] achieved a 55.5% increase in recall and a 32.6% improvement in the F1 score over traditional pre-train-and-fine-tuning methods by employing a training-free entity extraction filter and self-fusion strategy.

In the urban computing domain, SIGMUS, introduced by Wang and Srivastava [12], leverages LLMs to produce world knowledge to identify the relationships between incidents across multimodal urban data streams, including CCTV imagery, air quality measurements, weather data, and news reports, without relying on human-encoded rules. Similarly, Antonov and Tupikina [11] proposed a knowledge-based framework that integrates LLM-assisted extraction with temporal knowledge graphs that incorporate geocoded toponyms for urban event detection. However, a critical limitation that has been consistently documented across LLM applications is the hallucination problem: the generation of fluent but factually incorrect content [13], [14]. Studies have shown that even with zero-temperature settings, LLMs can exhibit significant hallucinations in named entity extraction tasks, although fine-tuning can substantially reduce hallucination rates [13]. Creo et al. [15] reported similar findings for SemEval-2025, indicating that hallucination detection and mitigation remain as active research challenges.

These observations motivate the hybrid extraction architecture adopted in our framework: fine-tuned models provide high-precision, reproducible extraction for well-defined entity types, whereas an LLM-based verifier supplies complementary semantic coverage for edge cases, with a confidence calibration mechanism that explicitly accounts for the hallucination risk inherent in LLM outputs. Recent studies on hallucination detection and mitigation services [33] and the impact of fine-tuning strategies on hallucination rates in named entity extraction [41] further support the importance of verification mechanisms in LLM-based extraction pipelines.

D. NEURO-SYMBOLIC REASONING OVER KNOWLEDGE GRAPHS

The emerging paradigm of neurosymbolic AI combines the pattern recognition capabilities of neural networks with the logical rigor and interpretability of symbolic reasoning [16], [17]. Recent studies have proposed unified frameworks that tightly couple neural and symbolic components for KG reasoning [34]. In the context of knowledge graphs, neurosymbolic methods have been successfully applied to link prediction, entity classification, and question-answering tasks [18], [19]. Bhoite [28] proposed a hybrid neurosymbolic

reasoning framework for enterprise KGs that integrates graph neural networks for representation learning with symbolic rule-based reasoning for logical inference, achieving 15–23% accuracy improvements while reducing query response time by 40%.

Several recent approaches have explored tighter couplings between neural and symbolic components. Boulakbech and Wannous [29] introduced a context-aware hybrid neurosymbolic method for KG enrichment that combined LLM-based language understanding with formal semantic reasoning, thereby supporting contextual triple generation, semantic alignment, constraint validation, and dynamic ontology evolution. Yu et al. [18] presented ONSEP, an online neural-symbolic framework for event prediction that leverages LLMs to generate structured event representations for downstream symbolic reasoning. Fu et al. [19] proposed NeuroSymActive, a differentiable neural-symbolic reasoning system with active exploration for KG question answering, demonstrating that the integration of rule-based reasoning with neural learning can significantly improve the performance of complex multi-hop queries.

The teacher-student knowledge distillation paradigm has proven particularly effective for transferring symbolic knowledge to neural architectures. In the graph learning domain, graph attention networks have demonstrated strong capabilities in KG reasoning tasks by adaptively weighting neighboring nodes through attention mechanisms that, when properly trained, approximate the behavior of rule-based inference while maintaining generalization to unseen configurations [20], [21]. Recent advances include hierarchical graph self-attention architectures [35], grid calculations combined with GAT for geographic KG reasoning [36], and relation-aware spectral decoupling attention networks [37], all of which demonstrate the growing sophistication of attention-based KG reasoning methods. However, the application of neurosymbolic reasoning to urban event management, where expert-defined rules governing risk causality and departmental jurisdiction must operate over noisy colloquial complaint narratives, represents a gap in the literature that this study aims to fill.

Table 1 summarizes the key characteristics of representative prior studies across the four reviewed areas, highlighting the integration of ST-CKG and the extension of existing approaches.

III. METHODOLOGY

In this section, we present the detailed design of the proposed ST-CKG framework. Before proceeding to the technical details, we clarify the scope of “causal” as used in this work. In the statistical causal inference literature, causality is formally defined through interventions and counterfactuals within the framework of structural causal models. The present work does not perform counterfactual reasoning or estimate average treatment effects. Rather, we adopt the risk causal chain perspective established in the community governance literature, where causality is encoded through expert-defined

TABLE 1. Comparison of representative prior work and ST-CKG advancements.

Research Area	Representative Work	Key Limitation	ST-CKG Advancement
KG for Smart Cities	Wang et al. [22], Rajan et al. [7]	Static representation; limited to physical infrastructure	Causal risk chain modeling; dynamic citizen complaint integration
Spatio-Temporal KG	Jin et al. [6], Han et al. [25]	Focus on traffic/environment; no risk causal semantics	Spatial proximity and temporal causal hyperedges for risk propagation
LLM for KG Construction	Wang & Srivastava [12], Sun et al. [27]	Hallucination risk; lack of confidence calibration	Hybrid extraction with LLM verifier and confidence scoring
Neuro-Symbolic Reasoning	Bhoite [28], Fu et al. [19]	General-purpose; not applied to urban governance	Rule distillation into GAT for urgency assessment and decision support

logical rules that specify deterministic relationships among risk factors: an entity exposed to a hazard may trigger a specific risk, which, when activated by an event, may lead to a particular consequence. This rule-based causal modeling, formalized in the Semantic Web Rule Language (SWRL), provides the inductive bias that guides the neuro-symbolic reasoning engine. We use the term “causal knowledge graph” to emphasize that our graph encodes directional, asymmetric cause–effect relationships, as opposed to symmetric co-occurrence patterns captured by conventional knowledge graphs. Extending this framework to support data-driven counterfactual reasoning is discussed as a future direction in Section V-E.

The framework is organized as a three-phase pipeline that progressively transforms unstructured citizen complaint narratives into structured, actionable governance decisions. Phase 1 performs hybrid information extraction with confidence calibration, combining fine-tuned domain-specific models with an LLM-based soft verifier. Phase 2 constructs a spatio-temporal causal hypergraph that encodes risk entities, their causal relationships, and spatio-temporal coupling effects through specialized hyperedge types. Phase 3 employs a neuro-symbolic reasoning engine that distills expert-defined symbolic rules into a graph attention network, enabling interpretable inference that generalizes beyond the coverage of handcrafted rules.

A. FRAMEWORK OVERVIEW

The ST-CKG architecture adopts a modular, sequentially composed design, wherein the output of each phase serves as the input to the subsequent phase. Fig. 1 presents the end-to-end system architecture, illustrating the complete data flow from the raw textual complaints to the final decision outputs.

The three phases are tightly integrated, yet functionally decoupled, allowing each component to be independently

evaluated, upgraded, or adapted to different urban governance contexts. Phase 1 received the raw complaint text and produced a set of structured tuples formally defined as (1):

$$\mathcal{T} = \{(e, h, r, v, c) \mid e \in \mathcal{E}_{\text{entity}}, h \in \mathcal{E}_{\text{hazard}}, r \in \mathcal{E}_{\text{risk}}, v \in \mathcal{E}_{\text{event}}, c \in [0, 1]\} \quad (1)$$

where each tuple has a confidence score c that quantifies the reliability of the extracted information. These tuples are passed to Phase 2, where they are mapped onto the Community Governance Ontology (CGO) and organized into a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ whose incidence matrix $\mathbf{H} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{E}|}$ encodes both standard binary relations and higher-order spatio-temporal hyperedges. Phase 3 takes this hypergraph as input, uses symbolic rules formalized in SWRL to generate teacher labels for urgency and departmental responsibility, and trains a Hypergraph Attention Network(HyperGAT) student model to perform soft, uncertainty-weighted inference. The final outputs include multiclass urgency predictions, recommended responsible departments, collaboration networks, and attention weight matrices that provide human-interpretable evidence for each decision.

The following subsections elaborate on each phase in detail: Section III-B formalizes the hybrid extraction problem and describes the fine-tuning, LLM verification, and confidence-fusion procedures. Section III-C presents the spatio-temporal causal hypergraph construction, including the ontology mapping scheme and mathematical formulation of the spatial and temporal hyperedges. Section III-D details the neuro-symbolic reasoning engine, covering the teacher-student distillation paradigm, HyperGAT architecture, weighted training objective, and attention-based interpretability mechanism.

B. PHASE 1: LLM-ENHANCED HYBRID INFORMATION EXTRACTION

The first phase of the ST-CKG framework addressed the fundamental challenge of transforming unstructured, colloquial citizen complaint narratives into structured knowledge tuples with quantified reliability estimates. This phase operates based on the principle that fine-tuned domain-specific and large-language models exhibit complementary strengths. The former offers high precision, controllability, and reproducibility for well-defined entity types within their training distribution, whereas the latter provides broader semantic coverage for edge cases, informal expressions, and implicit relations that deviate from the supervised training patterns. The core technical challenge lies in effectively combining their outputs while explicitly accounting for the hallucination risk inherent in the LLM generation.

1) PROBLEM FORMULATION

Let $\mathcal{X}$ denote the space of citizen complaint texts, where each instance $x \in \mathcal{X}$ is a free-form natural language narrative describing an urban incident, such as “Water seeps from the

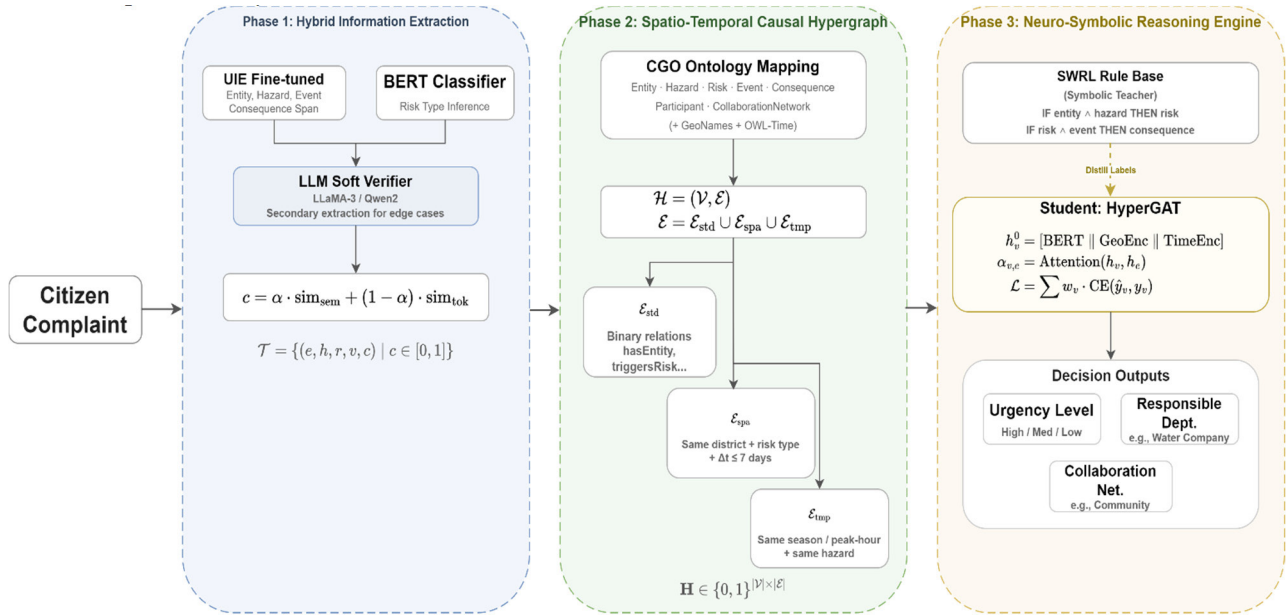


FIGURE 1. Overall architecture of the proposed ST-CKG framework, illustrating the three-phase pipeline: (Phase 1) hybrid information extraction with LLM-based soft verification and confidence calibration, (Phase 2) spatio-temporal causal hypergraph construction with three hyperedge categories, and (Phase 3) neuro-symbolic reasoning via rule distillation into a HyperGAT student model. The framework takes unstructured citizen complaint text as input and produces urgency assessments, responsible department assignments, collaboration network recommendations, and interpretable attention visualizations.

well in winter; when temperatures drop, ice forms on the road and pedestrians may fall.” The extraction objective is to produce a structured representation $\mathcal{T}(x) = \{(e, h, r, v, c)\}$ where $e \in \mathcal{E}_{\text{entity}}$ denotes the tangible asset or physical object involved (e.g., “water well”), $h \in \mathcal{E}_{\text{hazard}}$ denotes the latent dangerous condition (e.g., “water seepage”), $r \in \mathcal{E}_{\text{risk}}$ denotes the inferred risk type (e.g., “daily life disruption risk”), v optionally captures the trigger event or resulting consequence (e.g., “ice formation,” “pedestrian fall”), and $c \in [0, 1]$ is a calibrated confidence score quantifying the reliability of the extracted tuple.

The five extraction targets were categorized into two groups based on their surface form detectability. Explicit fields—entity e , hazard h , event, and consequence v —correspond to directly observable text spans and are amenable to sequence labeling approaches. Implicit fields—risk type r —require contextual inference beyond surface patterns and are formulated as text-classification tasks. This decomposition informs the design of the backbone extraction models.

2) BACKBONE MODEL FINE-TUNING

Following the methodology established in the community governance knowledge graph construction literature [23], we adopted a dual-model backbone comprising a UIE span for explicit field extraction, and BERT for implicit risk classification.

For explicit field extraction, we fine-tuned the Paddle-UIE model using annotated citizen-complaint data. The UIE employs a unified structured extraction paradigm that encodes extraction schemas and generates target structures

using sequence-to-sequence architecture [4]. The model was trained to identify text spans corresponding to entity, hazard, event, and consequence categories. Fine-tuning was performed with a learning rate $\eta = 1 \times 10^{-5}$, a batch size of 16, and 20 training epochs, following a protocol that achieved optimal performance on analogous community governance corpora. The fine-tuned model achieved a precision of 0.729, recall of 0.620, and F1-score of 0.670 for held-out evaluation data.

For implicit risk classification, we fine-tuned a BERT-based model to classify each complaint text into one of four risk categories: fire safety, daily life disruption, regulation violation, and conflict-prone [2]. The BERT model processes the full complaint text through its bidirectional transformer encoder and the final hidden state of the [CLS] token is passed through a linear classification head. The training used identical hyperparameters ($\eta = 1 \times 10^{-5}$, batch size = 16, and 20 epochs), yielding a classification accuracy of 0.8145. Per-category F1-scores were 0.795 (fire safety), 0.864 (daily life disruption), 0.666 (regulation violation), and 0.666 (conflict-prone).

3) LLM-BASED SOFT VERIFIER

While fine-tuned backbones provide reliable extraction within their training distribution, citizen complaints frequently contain colloquial expressions, domain-specific jargons, implicit descriptions, and irregular grammatical structures that deviate from supervised patterns. To complement the backbone with broader semantic coverage, we introduced an LLM-based soft verifier that performs independent secondary extraction on each complaint text.

The verifier was instantiated using an open-source instruction-tuned LLM (LLaMA-3-8B-Instruct or Qwen2-7B-Instruct) [9]. For each input text x , the LLM is prompted with a structured template that requests the extraction of entities, hazards, risk types, events, and consequences in a machine-parseable JSON format. The prompt design incorporates domain-specific definitions and few-shot examples to guide the model toward the expected output structure.

The LLM verifier produces a candidate structured output $\mathcal{T}_{\text{LLM}}(x)$ that may partially agree or disagree with backbone extraction $\mathcal{T}_{\text{BB}}(x)$. Critically, while LLM often captures semantically richer interpretations, particularly informal phrasings and implied information, it is susceptible to hallucinations, occasionally producing fluent but factually unsupported extractions [13], [14]. This motivated the confidence-aware fusion mechanism described below.

4) CONFIDENCE-AWARE FUSION STRATEGY

The fusion objective is to produce a consolidated set of structured tuples, each accompanied by a confidence score that reflects the degree of agreement between the backbone and LLM verifier. A high agreement indicates reliable extraction; disagreement signals either an edge case (where the LLM may be correct) or a potential LLM hallucination (where the backbone should be trusted).

For each overlapping extraction target, we computed two complementary similarity metrics between the backbone extraction and LLM candidate. (a) Semantic similarity: The text spans extracted by the two models are encoded using the same BERT-based encoder. The cosine similarity between the mean-pooled token embeddings is computed as given by (2):

$$\text{sim}_{\text{semantic}}(s_{\text{BB}}, s_{\text{LLM}}) = \frac{\mathbf{h}_{\text{BB}}^{\top} \mathbf{h}_{\text{LLM}}}{\|\mathbf{h}_{\text{BB}}\| \cdot \|\mathbf{h}_{\text{LLM}}\|} \quad (2)$$

where $\mathbf{h}_{\text{BB}}$ and $\mathbf{h}_{\text{LLM}}$ are the mean-pooled representations of the respective extracted text span. (b) Token overlap ratio: The character-level Jaccard similarity after whitespace normalization captures the surface-form agreement, as defined in (3):

$$\text{sim}_{\text{token}}(s_{\text{BB}}, s_{\text{LLM}}) = \frac{|s_{\text{BB}} \cap s_{\text{LLM}}|}{|s_{\text{BB}} \cup s_{\text{LLM}}|} \quad (3)$$

where $|\cdot|$ denotes character count after whitespace normalization.

The fused confidence score is a convex combination of these two metrics, as expressed in Equation (4).

$$c = \alpha \cdot \text{sim}_{\text{semantic}} + (1 - \alpha) \cdot \text{sim}_{\text{token}} \quad (4)$$

where $\alpha \in [0, 1]$ is a tunable weighting parameter, and through a grid search on a validation subset, we empirically set $\alpha = 0.6$, assigning a slightly higher weight to semantic similarity to accommodate synonymous expressions and paraphrasing.

When the LLM extraction exactly matches the backbone, both similarities approach 1 and the confidence score $c \approx 1$.

When the two models disagree substantially, both similarities drop and c approaches 0, flagging the tuple as unreliable.

A confidence threshold $\tau = 0.5$ is applied to determine the final extraction decision. Tuples with $c \geq \tau$ are accepted into the structured output; tuples with $c < \tau$ are flagged for optional human review but are still retained in the knowledge graph with their low-confidence labels to avoid information loss. The resulting confidence scores serve a dual purpose: they provide quality indicators for downstream human-in-the-loop verification workflows and supply sample weights for Phase 3 neurosymbolic reasoning training, in which low-confidence samples contribute proportionally less to the loss function.

The output of Phase 1 is a structured tuple set $\mathcal{T} = \{(e, h, r, v, c)\}$ for each complaint x . These tuples are passed to Phase 2, where they are mapped onto the Community Governance Ontology and organized into a spatiotemporal causal hypergraph. The BERT [CLS] token embeddings produced during the risk classification forward pass were cached and passed to Phase 3 for node feature initialization to avoid redundant recomputation and ensure semantic consistency across the pipeline.

C. PHASE 2: SPATIO-TEMPORAL CAUSAL HYPERGRAPH CONSTRUCTION

The second phase of the ST-CKG framework transforms the confidence-calibrated structured tuples produced in Phase 1 into a rich relational representation that explicitly encodes the causal, spatial, and temporal dependencies among urban risk events. Conventional knowledge graphs represent facts as binary triples (subject, predicate, and object), which suffice to model pairwise relationships, but fail to capture higher-order interactions, where multiple entities, hazards, events, and environmental conditions jointly determine risk propagation patterns. To overcome this limitation, we adopted a hypergraph formalism in which a single hyperedge can connect an arbitrary number of nodes, thereby naturally encoding the multiparty and multifactor nature of urban risk causality. This phase comprises three sequential steps: ontology-guided node instantiation, hypergraph construction with multitype hyperedges, and spatiotemporal augmentation.

1) ONTOLOGY-GUIDED NODE AND RELATION INSTANTIATION

The structured tuples $\mathcal{T}(x)$ extracted for each complaint x are first mapped onto the Community Governance Ontology (CGO), a domain-specific semantic model designed to capture the risk causal chain in urban governance contexts. CGO defines the following core classes: entities, hazards, risks, events, sequences, participants, and collaboration networks. Object properties connecting these classes encode the risk causal chain logic: `hasEntity` relates a complaint to the physical asset involved; `hasHazard` associates it with the identified latent danger; `triggersRisk` links an entity-hazard pair to the inferred risk type; `leadsToEvent` captures the

triggering condition under which the risk materializes; and causesConsequence models the adverse outcome.

Each field extracted from Phase 1 instantiates the corresponding node in the knowledge graph. Entity fields instantiate entity nodes; hazard fields instantiate hazard nodes; and risk-type classifications instantiate risk nodes. Event and Consequence fields when present in the extraction output instantiate event and consequence nodes, respectively. Decision-making knowledge, including responsible departments and collaboration networks extracted from the resolution records accompanying each complaint, instantiates participant and collaboration network nodes. The associated activities and resources are captured through additional activities and resource nodes, following the CGO schema.

Spatial and temporal attributes were associated with each complaint node through integration with the external geographic and temporal ontologies. Geolocation information (latitude, longitude, and administrative district) is populated via the GeoNames geographic ontology, whereas temporal attributes (timestamp, season, and time-of-day period) are encoded using the OWL-Time temporal ontology. Specifically, each complaint node is linked to a place instance (inherited from the GeoNames Feature) through an occurredAt property, and to a timeinstant instance through an occurredAtTime property. The Place instance is further connected to a street instance via an administratedBy relationship, thereby enabling hierarchical spatial reasoning. The TimeInstant instance is linked to a Season instance (spring, summer, autumn, winter) and TimePeriod instance (morning peak, midday, evening peak, night) through belongsToSeason and belongsToPeriod properties, respectively. This dual linkage forms the basis of the knowledge graph for both geographic and temporal reference systems, and is essential for the spatiotemporal hyperedge construction described below.

2) HYPERGRAPH FORMALIZATION

Building upon the instantiated nodes and binary relations, we constructed a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ that serves as the structural substrate for the neurosymbolic reasoning engine in Phase 3. A hypergraph extends the standard graph formalism by allowing each hyperedge $e \in \mathcal{E}$ to connect an arbitrary subset of nodes, rather than exactly two. This expressive power is particularly suited to modeling risk causal chains, where a single complaint event intrinsically involves an entity (the physical object), a hazard (the dangerous condition), a risk (the inferred threat), and potentially an event trigger and consequence, all of which should be jointly associated as a cohesive unit.

Formally, the hypergraph is defined as follows. The node set $\mathcal{V}$ comprises all instantiated CGO entities, as expressed in (5):

$$\begin{aligned} \mathcal{V} = & \mathcal{V}_{\text{Entity}} \cup \mathcal{V}_{\text{Hazard}} \cup \mathcal{V}_{\text{Risk}} \cup \mathcal{V}_{\text{Event}} \cup \mathcal{V}_{\text{Consequence}} \\ & \cup \mathcal{V}_{\text{Complaint}} \cup \mathcal{V}_{\text{Participant}} \cup \mathcal{V}_{\text{CollaborationNetwork}} \\ & \cup \mathcal{V}_{\text{Place}} \cup \mathcal{V}_{\text{Season}} \cup \mathcal{V}_{\text{TimePeriod}} \end{aligned} \quad (5)$$

The hyperedge set $\mathcal{E}$ is the union of three disjoint categories, given by (6):

$$\mathcal{E} = \mathcal{E}_{\text{standard}} \cup \mathcal{E}_{\text{spatial}} \cup \mathcal{E}_{\text{temporal}} \quad (6)$$

where each category captures a distinct dimension of the relational structure.

The incidence matrix $\mathbf{H} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{E}|}$ encodes the hypergraph topology, with $\mathbf{H}_{v,e} = 1$ if node v participates in hyperedge e , and 0 otherwise. From this matrix, we derive the node degree $d(v) = \sum_{e \in \mathcal{E}} \mathbf{H}_{v,e}$ and hyperedge degree $\delta(e) = \sum_{v \in \mathcal{V}} \mathbf{H}_{v,e}$, which quantify the connectivity of the individual nodes and size of each hyperedge, respectively. For the purpose of the graph neural network computation in Phase 3, the hypergraph Laplacian is approximated by the normalized adjacency matrix given by (7):

$$\mathbf{A}_{\text{hyper}} = \mathbf{H} \mathbf{W}_e \mathbf{H}^T \quad (7)$$

where $\mathbf{W}_e = \text{diag}(w(e_1), w(e_2), \dots, w(e_{|\mathcal{E}|}))$ is a diagonal matrix of hyperedge weights. We initialized all hyperedge weights to unity and allowed the attention mechanism in Phase 3 to learn adaptive weighting.

3) MULTI-TYPE HYPEREDGE CONSTRUCTION

The three hyperedge categories are constructed according to distinct semantic rules, each contributing a complementary perspective to the risk propagation dynamics.

Standard hyperedges ($\mathcal{E}_{\text{standard}}$) encode the fundamental risk causal chains within each complaint. For a given complaint node $v_{\text{complaint}}$, if Phase 1 extraction produces entities e, h, r, t , and c , a single standard hyperedge is created that connects all these nodes: $\{v_{\text{complaint}}, e, h, r, t\}$. This hyperedge captures the co-occurrence of the risk factors within a single incident. In cases in which the extraction does not contain an explicit event or consequence field, a hyperedge is formed over the available subset of nodes. Standard hyperedges constitute the base layer of the hypergraph, and preserve the complete risk profile for each complaint.

Spatial proximity hyperedges ($\mathcal{E}_{\text{spatial}}$) are designed to capture the geographically correlated propagation of urban risks. For any two distinct complaint nodes v_i and v_j with associated risk types r_i and r_j , we created a spatial proximity hyperedge if the following conditions were satisfied: (a) both complaints occurred within the same administrative district, as determined by the administratedBy property linking their respective place nodes to an identical street instance; (b) the two complaints shared the same inferred risk type, that is, $r_i = r_j$; and (c) the temporal interval between their occurrence timestamps did not exceed a window of $\Delta t = 7$ days. When these conditions hold, the spatial hyperedge connects $\{v_i, v_j, r_i\} \cup \mathcal{N}_{\text{shared}}(v_i, v_j)$, where $\mathcal{N}_{\text{shared}}$ denotes the set of entities or hazard nodes appearing in both the complaints. The rationale behind this construction is that multiple complaints of the same risk type, occurring in close spatial and temporal proximity, often indicate an underlying systemic hazard. For example, multiple reports of exposed electrical wiring across adjacent residential compounds within a week may

signify a broader infrastructure degradation issue requiring coordinated intervention. By encoding such spatial clustering explicitly as hyperedges, we provide a reasoning engine with structural signals that facilitates the detection of emerging risk hotspots.

Temporal causal hyperedges ($\mathcal{E}_{\text{temporal}}$) were designed to capture recurrent risk patterns that exhibit seasonal or diurnal periodicity in urban environments. For any two distinct complaint nodes v_i and v_j , a temporal causal hyperedge is created if (a) both complaints are linked to the same hazard type h (e.g., both involve “water seepage”); (b) both complaints share the same event trigger type t (e.g., both involve “low-temperature ice formation”); and (c) their occurrence timestamps belong to the same temporal category, defined as membership in the identical season instance (e.g., both occur in winter) or an identical time period instance (e.g., both occur during the morning peak). The temporal hyperedge connects $\{v_i, v_j, h, t\} \cup \mathcal{S}$, where $\mathcal{S}$ denotes a shared season or a period node. This construction explicitly encodes the learned domain knowledge regarding urban risk periodicity: certain hazards manifest predominantly during specific seasons (e.g., water pipe bursts in freezing winter conditions and fire hazards during dry summer months), and certain risk events are concentrated during peak commuting hours (e.g., road obstruction complaints and traffic signal malfunctions). By instantiating these temporal associations as a hypergraph topology, the framework enables the reasoning engine to anticipate risk recurrence based on calendar and clock contexts, which is essential for proactive governance strategies, such as preemptive resource allocation before known high-risk seasons.

The incidence matrix H produced in this construction phase, together with the three categories of hyperedges, served as the structured input for phase 3. The hypergraph representation unifies within a single mathematical object the diverse relational signals—individual risk profiles, spatial risk clustering, and temporal risk recurrence—that collectively determine the urgency and optimal response strategy for each citizen complaint.

D. PHASE 3: NEURO-SYMBOLIC REASONING ENGINE

The third phase of the ST-CKG framework addresses the core inferential challenges in urban event management. Given a citizen complaint represented as a subgraph within the spatiotemporal causal hypergraph, how can the system accurately assess urgency, assign responsible departments, and recommend collaboration networks in a manner that is consistent with established governance rules and robust to the noise, incompleteness, and novelty inherent in real-world complaint narratives? Purely symbolic approaches, such as the SWRL rule bases employed in prior community governance knowledge graphs, offer precision and interpretability, but fail catastrophically when confronted with cases that fall outside their predefined rule coverage. Conversely, purely neural approaches can generalize to unseen patterns, but lack

logical constraints that ensure compliance with regulatory frameworks and departmental jurisdiction boundaries. The proposed neuro-symbolic reasoning engine bridges this gap through a teacher-student knowledge distillation paradigm, which serves as a teacher that generates high-quality labels for cases within their coverage, while a HyperGAT student learns to reproduce and generalize these decisions, producing soft, uncertainty-aware predictions for all cases.

1) SYMBOLIC TEACHER GENERATION

The teacher component is constructed from the expert-defined risk causal chain and decision-making rules formalized in the Semantic Web Rule Language (SWRL). These rules, which were originally developed for the community governance knowledge graph, encode two categories of domain knowledge.

Risk causal analysis rules specify the logical conditions under which entity-hazard pairs imply specific risk types and under which risk-event conjunctions imply specific consequences. Representative examples include the entity-hazard inference rule, which states that if a complaint involves entity e and hazard h , and a known entity-hazard pattern $\langle e, h \rangle \rightarrow r$ exists in the knowledge base, then the risk type r can be inferred; and the event-risk inference rule, which states that if risk r coexists with event trigger t , and a known event-risk pattern $\langle r, t \rangle \rightarrow c$ exists, then consequence c can be inferred.

The decision-making rules encode the mapping of risk types and geographic jurisdictions into responsible departments and collaborative networks. For instance, the participant assignment rule states that if a complaint involves risk type r and occurs at a location administered by street s , and department d has jurisdiction over s with a responsibility type that covers r , then d is assigned as the primary responsible participant. The collaboration network rule further specifies that if department d is assigned and collaboration network n provides supporting responsibility for d 's responsibility type and has jurisdiction over the complaint location, n is recommended as the collaboration network.

Urgency assessment rules combine risk types in a spatiotemporal context. A complaint involving a “daily life disruption” risk that occurred during the morning or evening peak commuting period was assigned as high urgency. Similarly, a complaint involving “fire safety risk” that occurred during the summer season was assigned a high urgency. All other cases defaulted because of low urgency.

The teacher executed these rules over the training subset of the hypergraph. For each complaint node $v \in \mathcal{V}_{\text{Complaint}}$, the teacher attempts to match the antecedents of all the applicable rules to the incident hyperedges of the nodes and their connected nodes. When a complete rule match is found, the teacher produces a hard label vector $y_v = [y_v^{\text{urgency}}, y_v^{\text{participant}}, y_v^{\text{collaboration}}]$, where each component is a one-hot encoding over the respective label space (urgency: {high, medium, low}; participant: set of departmental categories; collaboration: set of collaboration network categories). Cases in which no rule chain fires completely are

left unlabeled and handled using the semi-supervised training strategy described below.

2) HyperGAT STUDENT ARCHITECTURE

The student model is a HyperGAT designed to perform node-level classification of the complaint nodes in the hypergraph. The architecture comprises of three stages: node feature initialization, hypergraph attention convolution, and multihead prediction.

Node feature initialization constructs a dense vector representation for each node by fusing the heterogeneous information sources. For complaint node v , the text of the original citizen complaint was encoded using the fine-tuned BERT model from Phase 1, and the final hidden state of the [CLS] token was considered the textual feature $\mathbf{h}_v^{\text{text}} \in \mathbb{R}^{768}$. The geographic coordinates (latitude and longitude) associated with the location node of the complaint are encoded using a sinusoidal position-encoding function, producing a 64-dimensional geographic feature vector $\mathbf{h}_v^{\text{geo}} \in \mathbb{R}^{64}$. Temporal information, comprising the month (one-hot, 12 dimensions), hour of day (one-hot, 24 dimensions), and season (one-hot, four dimensions), was concatenated into a 40-dimensional temporal feature vector $\mathbf{h}_v^{\text{time}} \in \mathbb{R}^{40}$. The three feature vectors were concatenated and projected onto a unified dimension $d = 256$ through a learned linear transformation, given by (8):

$$\mathbf{h}_v^{(0)} = \mathbf{W}_{\text{proj}}[\mathbf{h}_v^{\text{text}} \parallel \mathbf{h}_v^{\text{geo}} \parallel \mathbf{h}_v^{\text{time}}] + \mathbf{b}_{\text{proj}} \quad (8)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d \times (768+64+40)}$ and $\mathbf{b}_{\text{proj}} \in \mathbb{R}^d$. For non-complaint nodes (entities, hazards, risks, events, consequences, participants, and places), textual features were derived from their canonical name strings encoded through BERT, and geographic or temporal features were drawn from their associated attributes when available; otherwise, these dimensions were zero-padded. All node features were initialized using the same projection layer.

Hypergraph attention convolution performs two-layer message passing over a hypergraph structure. Let $\mathbf{h}_v^{(l)} \in \mathbb{R}^d$ denote the representation of node v in layer l ($l = 0, 1$). At each layer, the computation proceeds in two steps: hyperedge representation aggregation, and node representation update.

In the first step, for each hyperedge $e \in E$, its representation is computed by attending to all the nodes incident to e . The attention weight $\alpha_{v,e}^{(l)}$ quantifies the relevance of node v to hyperedge e at layer l , and is computed as (9):

$$\alpha_{v,e}^{(l)} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^{(l)\top}[\mathbf{W}^{(l)}\mathbf{h}_v^{(l)} \parallel \mathbf{W}^{(l)}\mathbf{h}_e^{(l)}]\right)\right)}{\sum_{u \in e} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^{(l)\top}[\mathbf{W}^{(l)}\mathbf{h}_u^{(l)} \parallel \mathbf{W}^{(l)}\mathbf{h}_e^{(l)}]\right)\right)} \quad (9)$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{d \times d}$ is a shared linear transformation, $\mathbf{a}^{(l)} \in \mathbb{R}^{2d}$ is the attention vector, $\parallel$ denotes vector concatenation, and $\mathbf{h}_e^{(l)}$ is the hyperedge representation from the previous layer (initialized as the mean of its incident node features

at $l = 0$). The hyperedge representation is then aggregated as the attention-weighted sum of its incident node features, as expressed in (10).

$$\mathbf{m}_e^{(l)} = \sum_{v \in e} \alpha_{v,e}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_v^{(l)} \quad (10)$$

In the second step, each node updates its representation by aggregating the representations of all hyperedges to which it belongs. Let $\mathcal{N}(v) = \{e \in \mathcal{E} : v \in e\}$ denote the set of hyperedges incident on the node v . The updated node representation is computed as (11):

$$\mathbf{h}_v^{(l+1)} = \sigma\left(\sum_{e \in \mathcal{N}(v)} \mathbf{m}_e^{(l)}\right) \quad (11)$$

where $\sigma(\cdot)$ denotes the ELU activation function. A residual connection was applied around each layer to stabilize the training: $\mathbf{h}_v^{(l+1)} = \mathbf{h}_v^{(l)} + \text{Dropout}(\mathbf{h}_v^{(l+1)})$, with a dropout rate of 0.3.

A multihead prediction was applied after the second convolution layer. For each complaint node v , the final representation $\mathbf{h}_v^{(2)}$ is passed through three separate linear classification heads: an urgency head producing a 3-way softmax distribution $\hat{y}_v^{\text{urgency}} \in \mathbb{R}^3$; a participant head producing an N_p -way softmax distribution $\hat{y}_v^{\text{participant}} \in \mathbb{R}^{N_p}$, where N_p is the number of distinct departmental categories; and a collaboration head producing an N_c -way softmax distribution $\hat{y}_v^{\text{collaboration}} \in \mathbb{R}^{N_c}$, where N_c is the number of distinct collaboration network categories.

3) TRAINING OBJECTIVE AND STRATEGY

The training objective combines supervised learning from teacher labels with semi-supervised regularization over unlabeled nodes.

The primary supervision signal was provided by teacher-labelled complaint nodes. For a labeled node v with teacher label $\mathbf{y}_v$ and student prediction $\hat{\mathbf{y}}_v$, the per-task cross-entropy loss is weighted by the confidence score c_v produced by Phase 1, as given by (12).

$$\mathcal{L}_{\text{sup}} = -\frac{1}{|\mathcal{V}_{\text{labeled}}|} \sum_{v \in \mathcal{V}_{\text{labeled}}} c_v \sum_{\text{task}} \sum_{k=1}^{K_{\text{task}}} y_{v,k}^{\text{task}} \log \hat{y}_{v,k}^{\text{task}} \quad (12)$$

where $\mathcal{V}_{\text{labeled}}$ is the set of complaint nodes for which the teacher successfully generated complete labels and K_{task} is the number of classes for each task. Confidence weight c_v ensures that predictions for cases with high-quality extractions are prioritized, whereas low-confidence cases contribute less to the gradient. For complaint nodes that fall outside the teacher's rule coverage (i.e., $\mathcal{V}_{\text{unlabeled}}$), we apply an entropy minimization regularization term defined in (13):

$$\mathcal{L}_{\text{ent}} = -\frac{\lambda}{|\mathcal{V}_{\text{unlabeled}}|} \sum_{v \in \mathcal{V}_{\text{unlabeled}}} \sum_{\text{task}} \sum_{k=1}^{K_{\text{task}}} \hat{y}_{v,k}^{\text{task}} \log \hat{y}_{v,k}^{\text{task}} \quad (13)$$

where λ is a balancing coefficient that is empirically set to 0.1. This term penalizes flat, uncertain prediction distributions and encourages students to make decisive inferences, even

in the absence of explicit teacher guidance. The total loss is $L = L_{\text{sup}} + L_{\text{ent}}$.

Training was performed using the AdamW optimizer with a learning rate 1×10^{-3} and a weight decay 5×10^{-4} . Owing to the manageable scale of the hypergraph (approximately 2,000 complaint nodes and several thousand auxiliary nodes), full-graph training was employed without mini-batch sampling. Early stopping was triggered when the validation accuracy of the urgency assessment did not improve for ten consecutive epochs. The model with the highest validation accuracy was used for the inference.

4) INTERPRETABILITY VIA ATTENTION VISUALIZATION

A key advantage of attention-based architecture is its inherent interpretability. For any prediction made by the student model, the attention weights $\alpha_{v,e}^{(l)}$ can be extracted and examined to identify which hyperedges and, by extension, which connected nodes contribute the most strongly to the decision.

For a complaint v classified as high urgency, the system retrieves the top- K hyperedges ranked by their cumulative attention weights across layers $\sum_l \alpha_{v,e}^{(l)}$. These hyperedges provide an evidential basis for determining urgency. A spatial hyperedge with high attention may connect v to three similar complaints in the same district within the previous week, thereby indicating an emerging risk cluster. A temporal causal hyperedge may link v to a known winter hazard pattern. Standard hyperedges with high attention highlight the specific entity-hazard-risk combination that drives the inference.

This interpretability mechanism is operationalized as part of the decision output. Each recommendation is accompanied by an evidence summary listing the historical cases and risk patterns that inform the prediction. For governance practitioners, this transparency enables auditability, builds trust in automated recommendations, and supports accountability by documenting the rationale for each dispatch decision.

The outputs of Phase 3 (urgency classification, departmental assignments, collaboration network recommendations, and attention-based evidence summaries) constituted the final deliverables of the ST-CKG framework. These structured decision outputs are presented through a visualization interface that enables human operators to review, validate, and when necessary, override the system's recommendations, thereby maintaining human oversight while benefiting from automated analytical support.

IV. EXPERIMENTS

This section presents a comprehensive empirical evaluation of the proposed ST-CKG framework. First, we describe the experimental setup including the dataset, baseline methods, evaluation metrics, and implementation details. We then report the results of three core components: hybrid information extraction performance, neurosymbolic reasoning accuracy, and a series of ablation and case studies that isolate the contribution of each design choice.

A. EXPERIMENTAL SETUP

1) DATASET

Experiments were conducted on a real-world dataset derived from the immediate response to the complaint citizen service platform of District H, City B, spanning the full calendar year of 2021 [23]. The original dataset contained 2,007 anonymized citizen complaint records, each comprising 12 fields, including complaint identifiers, submission timestamps, geocoded locations, thematic category codes, detailed free-text complaint narratives, official resolution opinions, and organizational identifiers of the receiving and handling departments. After applying the data extraction and entity alignment procedure described in Phase 1, the resulting knowledge graph contains the instances summarized in Table 2. Spatiotemporal enrichment links each complaint to geocoded places and temporal category nodes through integration with the GeoNames and OWL-Time ontologies. The dataset was chronologically partitioned into training (January–September 2021, 1,405 complaints), validation (October–November 2021, 301 complaints), and testing (December 2021, 301 complaints) subsets. This temporal split reflects realistic deployment conditions, in which the models are trained on historical data and evaluated in the future with unseen complaints. For the reasoning tasks, we further partitioned the test set into rule-covered (217 complaints) and rule-uncovered (84 complaints) subsets, based on whether the symbolic teacher generated complete labels. The ground-truth labels for both subsets were derived from the official resolution records accompanying each complaint in the platform database. These records document the actual responsible department assigned, the collaboration network engaged, and the final resolution outcome (e.g., on-site inspection, demolition order, infrastructure repair). The urgency level for each complaint was determined from the time-to-resolution recorded in the dataset, which resolved within 24 h of filing as high urgency, those resolved within 24 to 72 h as medium urgency, and those requiring more than 72 h as low urgency. This labeling protocol reflects the operational service-level objectives of the municipal complaint-handling system, and was validated through consultation with domain experts familiar with the platform's governance workflows. Consequently, all 301 test complaints, whether covered or uncovered by the SWRL rule base, verified ground-truth annotations that enable rigorous evaluation of both the symbolic teacher and the neurosymbolic student.

2) BASELINES

We compared the proposed ST-CKG framework with a comprehensive set of baseline methods, as summarized in Table 3. To strengthen the comparative evaluation, we added several stronger baselines, including two state-of-the-art large language models (GPT-4o and DeepSeek-V2) for extraction tasks, a CompGCN model for knowledge graph reasoning, and a fine-tuned BERT classifier for direct urgency prediction.

TABLE 2. Statistics of the constructed knowledge graph.

Category	Type	Count
Causal Analysis	Entity	722
	Hazard	464
	Risk	4
	Event	16
Decision-Making	Consequence	100
	Participant (Department)	188
	Collaboration Network	113
	Participant Activity	346
	Collaboration Activity	80
Spatio-Temporal	Resource	3
	Geocoded Place Instances	2,824
Total Complaints		2,007

TABLE 3. Summary of baseline methods.

Category	Method	Description
Extraction	Original UIE	Paddle-UIE without domain fine-tuning, zero-shot
	ONEEKE	Open-domain nested event knowledge extraction
	Qwen-7B-Chat	Instruction-tuned LLM, zero-shot structured prompt
	LLaMA-3-8B-Instruct	Instruction-tuned LLM, zero-shot structured prompt
	GPT-4o (zero-shot)	Commercial LLM, zero-shot structured prompt
	DeepSeek-V2 (zero-shot)	Open-source MoE LLM (236B total, 21B activated), zero-shot prompt
	Backbone	Fine-tuned UIE + BERT (Phase 1, no LLM verifier)
	Hybrid (Ours)	Full Phase 1: Backbone + LLM verifier + fusion
Reasoning	Rule-only	SWRL symbolic rules executed on KG
	BERT (fine-tuned)	BERT classifier trained on complaint text for urgency
	Standard GAT	Flat graph, no hyperedges, no rule distillation
	CompGCN	Composition-based GCN on the KG without hyperedges
	HyperGAT w/o KD	HyperGAT with uniform weights, no rule distillation
Ablation	Neuro-Symbolic (Ours)	Full Phase 3: HyperGAT + rule distillation + confidence weights
	ST-CKG w/o LLM Verifier	Remove LLM soft verifier from Phase 1
	ST-CKG w/o Spatial Hyperedges	Remove E_{spatial} from Phase 2
	ST-CKG w/o Temporal Hyperedges	Remove E_{temporal} from Phase 2
	ST-CKG w/o Confidence Weighting	Remove c_v weighting from Phase 3 loss

3) EVALUATION METRICS

The metrics employed for each evaluation dimension are defined in Table 4.

TABLE 4. Evaluation metrics by task.

Task	Metric	Formula / Description
Explicit Field Extraction	Precision, Recall, F1	Span-level exact match; micro-averaged across four fields weighted by per-field test sample count
Risk Classification	Accuracy, Per-class F1	Standard multi-class metrics
Urgency Assessment	Accuracy, Macro-F1	3-class: high / medium / low
Department Assignment	Top-1 Accuracy	Correct department among N_p
Collaboration Recommendation	Top-1 Accuracy	Correct network among N_c
Efficiency	Inference Latency	Average seconds per complaint
	LLM API Calls	Number of calls per complaint

TABLE 5. Implementation and hyperparameter configuration.

Component	Parameter	Value
Hardware	CPU	Dual Intel Xeon Gold 6248R
	GPU	NVIDIA A100 40 GB
Phase 1 Fine-Tuning	Learning rate	1×10^{-5}
	Batch size	16
	Epochs	20
Phase 1 Fusion	α (semantic weight)	0.6
	τ (confidence threshold)	0.5
Phase 3 HyperGAT	Hidden dimension d	256
	Number of layers	2
	Dropout rate	0.3
	Learning rate	1×10^{-3}
	Weight decay	5×10^{-4}
	Max epochs	200
	Early stopping patience	10
Software Libraries	λ (entropy regularization)	0.1
	NLP Framework	PaddleNLP, HuggingFace Transformers
	KG Storage	GraphDB 10.1 Free Edition
	GNN Framework	PyTorch Geometric
	LLM Inference	vLLM

4) IMPLEMENTATION DETAILS: THE HARDWARE, SOFTWARE, AND HYPERPARAMETER CONFIGURATIONS ARE SUMMARIZED IN TABLE 5

All the hyperparameters were determined through a grid search of the validation subset. Unless otherwise specified, all experimental results reported in this section are the mean of three independent runs with different random seeds. The sensitivity of the framework to the key parameters α , τ , and λ is systematically evaluated in Section IV-D.

B. INFORMATION EXTRACTION PERFORMANCE

The extraction capabilities of the proposed hybrid pipeline are evaluated against those of the baselines described in

Section IV-A. The evaluation is organized along two dimensions: extraction of explicit fields (entity, hazard, event, and consequence spans), where surface-form text patterns are directly observable; and classification of the implicit risk-type field, which requires contextual inference beyond lexical cues.

1) EXPLICIT FIELD EXTRACTION

Table 6 reports the span-level F1-scores for extracting entities, hazards, events, and consequence fields from citizen complaint texts. The proposed hybrid method achieved a micro-averaged F1-score of 0.703, outperforming all baselines by a margin of at least 3.3 percentage points.

TABLE 6. Explicit field extraction performance (Span-level F1).

Method	Entity	Hazard	Event	Consequence	Micro-Avg
Original UIE	0.083	0.052	0.041	0.064	0.060
ONEEKE	0.371	0.348	0.322	0.361	0.353
Qwen-7B-Chat (zero-shot)	0.468	0.451	0.433	0.470	0.456
LLaMA-3-8B-Instruct	0.455	0.438	0.419	0.453	0.442
GPT-4o(zero-shot)	0.487	0.473	0.455	0.489	0.476
DeepSeek-V2 (zero-shot)	0.502	0.488	0.469	0.501	0.490
Backbone	0.729	0.710	0.652	0.714	0.670
Hybrid (Backbone+ LLM Verifier)	0.762	0.743	0.681	0.740	0.703

Note: Micro-Avg is computed as the weighted average of the four per-field F1-scores, where the weight for each field is proportional to the number of test samples containing that field. Event and Consequence fields have fewer annotated instances than Entity and Hazard fields in the dataset (as reported in Table 2). Consequently, the weighted Micro-Avg differs from the simple arithmetic mean of the four columns, particularly for the Backbone and Hybrid methods where per-field performance varies substantially. Regarding the zero-shot LLM baselines, the slightly higher scores of DeepSeek-V2 relative to GPT-4o are consistent with the model's design emphasis on Chinese-language understanding. DeepSeek-V2 adopts a Mixture-of-Experts architecture with 236B total parameters (21B activated per token) and has demonstrated performance comparable to GPT-4 on Chinese-centric benchmarks. The citizen complaint dataset used in this study is predominantly Chinese-language, which favors models with stronger Chinese linguistic capabilities. The performance gap between the two models is small (1–1.5 percentage points) and the core finding—that domain-specific fine-tuning substantially outperforms all zero-shot LLMs—remains robust regardless of the relative ordering between these two baselines.

The fine-tuned backbone provided substantial improvements over all zero-shot approaches; it achieved an F1-score of 0.670, representing a gain of 61.0 percentage points over the Original UIE (0.060) and 31.7 points over ONEEKE (0.353). This dramatic gap underscores the necessity of domain-specific fine-tuning for specialized vocabulary, non-standard grammar, and colloquial expressions characteristic of citizen complaints. The zero-shot LLM baselines performed significantly better than the zero-shot UIE: LLaMA-3-8B-Instruct achieved 0.442, and Qwen-7B-Chat achieved 0.456, demonstrating the intrinsic semantic understanding capacity of large-scale pre-trained models. However, both

LLM baselines remained approximately 21 points below the fine-tuned backbone, confirming that zero-shot prompting alone is insufficient for high-precision extraction in this specialized domain.

The newly added GPT-4o and DeepSeek-V2 baselines further confirmed this trend. Despite their larger scale and stronger zero-shot capabilities, they reach F1-scores of only 0.476 and 0.490, respectively, which are still over 18 points below the fine-tuned backbone. This reinforces the necessity of domain-specific fine-tuning, and validates our hybrid design.

The Hybrid method adds a further 3.3 percentage point gain over the backbone by incorporating an LLM soft verifier with confidence-aware fusion. Analysis of the per-field breakdown reveals that the improvement is most pronounced for the event field (2.9 point gain from 0.652 to 0.681) and the hazard field (3.3 point gain from 0.710 to 0.743). These fields frequently involve implicit and inferential descriptions of citizens' complaints. For instance, a hazard may be described indirectly ("water accumulates on the road surface after rain and becomes dangerously slippery") rather than through a standardized term, which broader semantic understanding successfully recovers cases in which the fine-tuned span extractor misses the target because of the distributional shift from its supervised training patterns. In contrast, the Entity field shows a more modest improvement (3.3 point gain from 0.729 to 0.762), reflecting the comparatively higher regularity of entity mentions (e.g., well-defined physical objects such as "manhole cover," "power line," and "building facade") for which the fine-tuned extractor is already well-optimized.

Fig. 2 presents the distribution of the confidence scores produced by the Phase 1 fusion mechanism across all extracted tuples. The scores were aggregated per complaint by averaging constituent tuples. Score range [0, 1] was discretized into ten equal-width bins of size 0.1. Approximately 72.4% of complaints received mean confidence scores above 0.7, indicating a strong agreement between the Backbone and the LLM verifiers. Further, 14.4% falls in the intermediate range of 0.3 to 0.7, representing cases where the two models partially agree or where one model's extraction is a subset of the others. The remaining 13.2% of complaints scored below 0.3, indicating flagging extraction with substantial disagreement that warrants human review. Manual inspection of 50 randomly sampled low-confidence complaints revealed three primary sources of disagreement: ambiguous pronoun references (42%), incomplete entity mentions (34%), and domain-specific abbreviations or local toponyms incorrectly interpreted by the LLM (24%). This analysis confirms that the confidence score provides a reliable mechanism for identifying extraction instances that would benefit from human oversight, thereby enabling targeted quality assurance in operational deployment.

2) RISK TYPE CLASSIFICATION

Table 7 presents the classification accuracy and per-class F1-scores for the implicit risk-type inference task. The

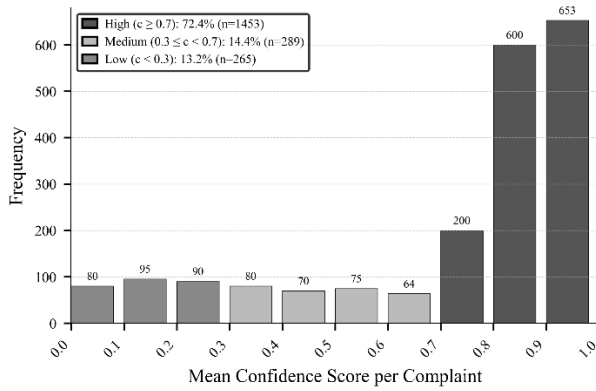


FIGURE 2. Distribution of mean confidence scores per complaint produced by the Phase 1 hybrid fusion mechanism. Scores are binned into ten equal-width intervals. The three regions correspond to high-confidence ($c \geq 0.7$, solid), medium-confidence ($0.3 \leq c < 0.7$, light), and low-confidence ($c < 0.3$, hatched) categories. The distribution is strongly right-skewed; low-confidence complaints (13.2%) are flagged for optional human review.

fine-tuned BERT Backbone achieved an accuracy of 0.815, with the strongest per-class performance on Daily Life Disruption (F1 = 0.864) and Fire Safety Risk (F1 = 0.795). The zero-shot LLM baselines performed substantially worse: LLaMA-3-8B-Instruct achieved 0.614 accuracy, whereas Qwen-7B-Chat achieved 0.593 accuracy with a gap exceeding 20 points. This underscores a critical finding: Risk classification requires deep domain-specific reasoning beyond general-purpose semantic understanding.

TABLE 7. Risk type classification performance.

Method	Accu racy	F1: Fire Safety	F1: Daily Life	F1: Reg. Violation	F1: Conflict
Qwen-7B-Chat	0.593	0.541	0.632	0.487	0.512
LLaMA-3-8B-Instruct	0.614	0.568	0.648	0.502	0.537
GPT-4o (zero-shot)	0.635	0.582	0.671	0.523	0.558
DeepSeek-V2 (zero-shot)	0.648	0.598	0.683	0.540	0.571
BERT (Fine-tuned Backbone)	0.815	0.795	0.864	0.666	0.666

The Hybrid method does not directly modify the risk classification output because it is an implicit inference task for which the fine-tuned BERT classifier serves as the primary model and the LLM verifier operates on explicit span fields only. The structured tuples forwarded to Phases 2 and 3 inherit the risk type from the BERT classifier, along with confidence-calibrated extractions.

3) EFFICIENCY

The average end-to-end extraction latency per complaint is 0.34 seconds for the Backbone and 0.53 seconds for the Hybrid configuration (+56%). This additional time is attributable to the single LLM inference call issued per complaint, which extracts all target fields simultaneously. Thus, the Hybrid design achieves a favorable precision–efficiency

tradeoff: a 4.9% relative improvement in extraction F1 at a modest latency increase that remains well within the budget of non-real-time complaint triage systems.

In summary, the Phase 1 experiments yielded three principal findings. First, domain-specific fine-tuning is indispensable, with the backbone outperforming all zero-shot approaches by margins that exceed 20 F1 points. Second, the LLM soft verifier with confidence-aware fusion provides complementary improvement, particularly for semantically complex fields, whereas calibrated scores enable targeted quality assurance. Third, the fine-tuned discriminative models substantially outperformed the zero-shot generative LLMs in implicit inference tasks, thereby validating the asymmetric hybrid design.

C. REASONING AND DECISION-MAKING PERFORMANCE

This section evaluates the inferential capabilities of the proposed neurosymbolic reasoning engine against the baselines described in Section IV-A. The evaluation encompassed three tasks: urgency assessment, responsible department assignment, and collaboration network recommendations. It was stratified by rule coverage to assess generalizations beyond symbolic teachers.

1) URGENCY ASSESSMENT

Table 8 presents the urgency classification performance. The proposed neurosymbolic method achieved an overall accuracy of 0.912, outperforming all the baselines with substantial margins. In the rule-covered subset, both rule-only and neurosymbolic attained perfect accuracy (1.000), which was expected because the teacher labels for these cases were deterministically generated from the SWRL rules executed by the rule-only baseline. Standard GAT and HyperGAT without knowledge distillation (w/o KD) achieved 0.816 and 0.853, respectively, on the rule-covered subset, demonstrating that purely data-driven approaches can approximate rule-based logic but lack the precision guarantees of symbolic reasoning.

TABLE 8. Urgency assessment performance.

Method	Rule-Covered (n=217)	Rule-Uncovered (n=84)	Overall (n=301)
Rule-only	1.000	0.000	0.721
BERT (fine-tuned)	0.746	0.423	0.656
Standard GAT	0.816	0.512	0.730
CompGCN	0.837	0.541	0.754
HyperGAT w/o KD	0.853	0.560	0.771
Neuro-Symbolic (Ours)	1.000	0.690	0.912

Critical differentiation emerges in a rule-uncovered subset. The Rule-only baseline fails completely (accuracy 0.000), because these cases do not match any of the SWRL rule antecedents. The standard GAT achieves only 0.512, and the HyperGAT without knowledge distillation reaches 0.560. The Neuro-Symbolic method attains 0.690 on this challenging subset, representing a relative improvement of 23.2% over the best neural-only baseline (HyperGAT w/o KD). This

result validates the central hypothesis of the teacher-student distillation paradigm: the HyperGAT student, trained with rule-derived supervision on covered cases and entropy regularization on uncovered cases, learns to generalize the underlying risk logic encoded in SWRL rules to structurally similar but rule-unmatched complaints. For example, a complaint involving “exposed gas pipe near barbecue stall during summer heatwave” may not trigger any existing rule, but the learned attention patterns connect it through spatial hyperedges to fire safety complaints in adjacent districts and through temporal hyperedges to prior summer-season fire risk events, enabling the student to infer high urgency despite the absence of an explicit teacher label.

2) DEPARTMENT ASSIGNMENT AND COLLABORATION RECOMMENDATION

Tables 9 and 10 report the accuracies of the responsible department assignments and collaboration network recommendations, respectively. The pattern observed in the urgency assessment was replicated across both the tasks. In the rule-covered subset, the rule-only and neurosymbolic baselines achieve perfect accuracy because department assignments and collaboration recommendations are deterministically encoded in decision-making SWRL rules. Neural baselines perform competitively but with residual errors that reflect the difficulty of precisely memorizing assignment rules from sparse supervision signals.

TABLE 9. Responsible department assignment accuracy.

Method	Rule-Covered	Rule-Uncovered	Overall
Rule-only	1.000	0.000	0.721
BERT (fine-tuned)	0.721	0.398	0.631
Standard GAT	0.793	0.476	0.704
CompGCN	0.812	0.508	0.728
HyperGAT w/o KD	0.834	0.524	0.749
Neuro-Symbolic (Ours)	1.000	0.643	0.902

TABLE 10. Collaboration network recommendation accuracy.

Method	Rule-Covered	Rule-Uncovered	Overall
Rule-only	1.000	0.000	0.721
BERT (fine-tuned)	0.698	0.375	0.607
Standard GAT	0.770	0.452	0.681
CompGCN	0.793	0.476	0.705
HyperGAT w/o KD	0.811	0.488	0.725
Neuro-Symbolic (Ours)	1.000	0.619	0.894

In the rule-uncovered subset, the neuro-symbolic method achieved a department assignment accuracy of 0.643 and collaboration network recommendation accuracy of 0.619, compared with 0.000 for rule-only and 0.524 and 0.488 for the strongest neural baseline (HyperGAT without KD). The performance gap is slightly wider for collaboration recommendations than for department assignments, reflecting the greater combinatorial complexity of collaboration network determination, which involves reasoning over both risk types and multi-departmental jurisdictional relationships.

The overall accuracy across all 301 test complaints was 0.902 for department assignments, and 0.894 for collaborative recommendations. These figures represent substantial improvements over the existing purely rule-based system deployed in the original community governance knowledge graph, which achieves 0.721 for both tasks. This figure exactly mirrors the proportion of rule-covered cases in the test set because the rule only contributes zero correct predictions on rule-uncovered cases.

3) GENERALIZATION ANALYSIS

To quantify the generalization gain attributable to the knowledge distillation strategy, we compared the neurosymbolic method against a variant that entirely omitted the rule-based teacher and trained HyperGAT solely on labeled cases with uniform weights. This ablated variant achieved 0.771 urgency, 0.749 department, and 0.725 collaboration accuracies, with gaps of 14.1, 15.3, and 16.9 percentage, respectively. The magnitude of these gaps demonstrates that the symbolic teacher provides a supervisory signal whose value extends well beyond the labeled cases themselves: by structuring the learned representation space around rule-derived decision boundaries, the teacher regularizes the student’s learning trajectory, preventing overfitting to surface-level spurious correlations in the limited labeled data and promoting the emergence of generalizable reasoning patterns.

A detailed error analysis of the rule-uncovered subset revealed that residual errors of the neurosymbolic method were concentrated in complaints with highly atypical risk combinations. For example, a complaint involving simultaneous fire safety and traffic disruption risks, which is a compound risk scenario that is not anticipated by the current SWRL rule base, was misclassified. Such cases highlight the need for incremental rule-based expansion guided by deployment feedback, which is a direction for future research.

In summary, Phase 3 experiments confirm that the neurosymbolic reasoning engine successfully bridges the gap between the precision of symbolic rules and the generalization capacity of graph neural networks. In rule-covered cases, the students faithfully reproduced their teacher decisions. In rule-uncovered cases, the student substantially outperformed both the purely symbolic and neural alternatives, demonstrating meaningful generalization to novel risk configurations.

4) CROSS-THEME GENERALIZATION

To further assess the robustness of ST-CKG beyond a single complaint category, we conducted a cross-theme evaluation using thematic labels provided in the dataset. The original dataset categorized complaints into multiple thematic types, including Urban Management, Public Safety, Environmental Protection, and Housing Disputes. We selected the Urban Management category (1,022 complaints) for training and held complaints from the Public Safety category (138 complaints) as the test set. The neuro-symbolic model trained

solely on Urban Management data was evaluated based on Public Safety complaints without any fine-tuning or parameter adjustment.

Table 11 reports the results. The model retains an overall urgency accuracy of 0.852 on Public Safety complaints, compared to 0.912 on the original Urban Management test set. The Rule-only baseline achieves only 0.637, reflecting the limited transferability of rules designed primarily for Urban Management scenarios. HyperGAT without knowledge distillation achieves 0.714. The modest 6.0 percentage point drop of the full Neuro-Symbolic model indicates that the learned risk causal patterns and spatio-temporal reasoning transfer reasonably across thematic categories, although a domain gap exists. The spatial and temporal hyperedges, which encode generic proximity and seasonal recurrence patterns, contribute to this cross-theme robustness, as they are not tied to any single complaint category. This experiment demonstrates that ST-CKG is not overfit to a single theme and supports its potential for broader urban governance applications.

TABLE 11. Cross-theme generalization performance (Train on Urban Management, Test on public safety).

Method	Urgency Accuracy
Rule-only	0.637
HyperGAT w/o KD	0.714
Neuro-Symbolic	0.852

5) CROSS-TEMPORAL GENERALIZATION ANALYSIS

To further evaluate the generalization capability of the ST-CKG under a temporal distribution shift, a cross-temporal experiment was conducted. The original dataset spans January to December 2021 and was chronologically split into training (January–September, 1,405 complaints) and testing (Dec, 301 complaints), as described in Section IV-A. While this split already respects temporal ordering, the training and test periods are adjacent within the same calendar year, and may share seasonal patterns. To simulate a more challenging scenario in which event patterns, policy priorities, and public concerns evolve over a longer time horizon, we constructed an alternative split: complaints from January to September 2021 remained the training set, while complaints from October to December 2021 (originally part of the validation set, 301 complaints) were held out as an unseen future test set. This introduces a temporal gap between training and evaluation, testing whether the model can be generalized to later periods with potentially shifted urgency distributions and departmental coordination patterns.

Table 12 reports the results. The full Neuro-Symbolic model achieves an urgency accuracy of 0.881 on the temporally shifted test set, representing a 3.1 percentage point decline from the 0.912 achieved on the adjacent test month (December 2021). The Rule-only baseline drops more sharply from 0.721 to 0.674, reflecting the aging of rules designed for earlier complaint patterns. HyperGAT without

knowledge distillation declines from 0.771 to 0.728. The modest degradation of the Neuro-Symbolic model relative to baselines demonstrates that the spatio-temporal hyperedges and rule distillation provide a stabilizing effect under temporal distribution shift: spatial hyperedges remain informative regardless of temporal drift, while the distilled rule logic captures invariant departmental jurisdiction constraints. The cross-temporal experiment complements the cross-theme evaluation in Section IV-C 4, jointly confirming that ST-CKG does not overfit to the specific temporal or thematic distribution of its training data.

TABLE 12. Cross-temporal generalization performance.

Method	Urgency Accuracy
Rule-only	0.674
HyperGAT w/o KD	0.728
Neuro-Symbolic	0.881

D. ABLATION AND ANALYSIS STUDY

To isolate the contribution of each architectural component and validate the design choices underlying the ST-CKG, we conducted a comprehensive ablation study. We systematically removed the LLM soft verifier, two categories of spatiotemporal hyperedges, and a rule distillation mechanism that measures the impact on both extraction quality and downstream reasoning performance. We also provide a comparative cost analysis of the proposed hybrid pipeline and the pure LLM-based extraction approach.

1) EFFECT OF LLM SOFT VERIFIER

Removing the LLM verifier reduces phase 1 to the fine-tuned backbone alone. As reported in Table 6, this causes the extraction micro-averaged F1-score to drop from 0.703 to 0.670, with the largest degradation occurring in semantically complex fields, such as hazards and events. The downstream consequences of the reasoning are equally measurable. When the HyperGAT student was trained on backbone-extracted tuples (without confidence calibration), the overall urgency assessment accuracy decreased from 0.912 to 0.874 and the department assignment accuracy decreased from 0.902 to 0.858 (Table 13). Performance erosion is particularly evident in the rule-uncovered subset, where the absence of calibrated confidence weights prevents the student from appropriately discounting noisy extractions, causing error propagation from phases 1 to 3.

2) EFFECTS OF SPATIOTEMPORAL HYPEREDGES

We evaluated three variants: (i) without spatial hyperedges (E_{spatial} -removed), (ii) without temporal hyperedges (E_{temporal} -removed), and (iii) without both. Because urgency assessment relies heavily on spatiotemporal context—identifying risk clusters in neighboring districts and recognizing seasonal recurrence patterns—we report recall for the high-urgency class, which is the most critical for operational prioritization. The full ST-CKG achieved a high-urgency

recall of 0.88. Removing spatial hyperedges reduced recall to 0.72, removing temporal hyperedges to 0.76, and removing both degraded recall to 0.65 (Table 14). These drops confirm that the spatial proximity and temporal recurrence patterns provide complementary signals that HyperGAT exploits to identify emerging risk hotspots.

TABLE 13. Impact of LLM verifier removal on reasoning accuracy.

Configuration	Urgency	Dept. Assign.	Collab. Rec.
Full ST-CKG (Hybrid extraction)	0.912	0.902	0.894
w/o LLM Verifier (Backbone extraction)	0.874	0.858	0.847

TABLE 14. High-urgency recall under spatio-temporal hyperedges ablation.

Configuration	High-Urgency Recall
Full ST-CKG (all hyperedges)	0.88
w/o E_{spatial}	0.72
w/o E_{temporal}	0.76
w/o both E_{spatial} and E_{temporal}	0.65

3) EFFECT OF RULE DISTILLATION

To assess the importance of the symbolic teacher, we compared the full neurosymbolic model with HyperGAT trained without knowledge distillation (i.e., supervised only on available labels with uniform weights and no entropy regularization). While the overall accuracy gap was substantial (0.771 vs. 0.912 for urgency), the most revealing analysis stratified department assignment accuracy according to risk category. Table 15 shows that the full model maintains strong performance across all four risk types, including the two low-frequency categories (Regulation Violation and Conflict-Prone), with accuracies of 0.86 and 0.83 respectively. In contrast, HyperGAT without knowledge distillation suffered catastrophic degradation in these infrequent categories, dropping to 0.52 and 0.49. The regularizing effect of the rule-derived teacher prevents overfitting to the dominant risk types and preserves the logical structure required for correct decision-making on rare events, a capability that is critical in real-world settings where high-impact risks, though infrequent, demand reliable responses.

TABLE 15. Department assignment accuracy by risk type.

Configuration	Fire Safety	Daily Life	Reg. Violation	Conflict
Neuro-Symbolic (Full)	0.93	0.94	0.86	0.83
HyperGAT w/o KD	0.85	0.88	0.52	0.49

4) COST ANALYSIS

The economic feasibility of large-scale deployment was assessed by comparing the per-complaint processing cost of the proposed hybrid pipeline against a pure LLM-based extraction baseline. The LLM verifier was assumed to be served on an NVIDIA A100 GPU with an on-demand

cloud price of \$2.50/hour. Phase 1 hybrid extraction takes 0.53 seconds per complaint (including one LLM forward pass), yielding a total GPU time of 0.30 hours for the full dataset of 2,007 complaints, which translates to a cost of \$0.75. Adding the amortized training cost of the backbone (approximately \$5 for one-time fine-tuning) results in a total of roughly \$5.75 for the entire dataset or \$0.0029 per complaint. In contrast, evaluating a commercial LLM API (e.g., GPT-4o at \$2.50 per million input tokens and \$10.00 per million output tokens, with an estimated 500 input tokens and 100 output tokens per complaint) incurs a cost of approximately \$0.00225 per call, which is comparable on a per-complaint basis. However, the pure LLM baseline delivered a substantially lower extraction accuracy (0.442 F1 vs. 0.703) and lacked calibrated confidence scores and controlled schema adherence for the hybrid design. Moreover, the hybrid pipeline can be further optimized through batching and model quantization. These figures demonstrate that the proposed framework achieves a favorable cost-accuracy trade-off, making it viable for sustained operation on municipal complaint volumes.

5) SENSITIVITY ANALYSIS OF KEY PARAMETERS

To assess the robustness of the ST-CKG framework with respect to its critical hyperparameters, we conduct a sensitivity analysis on three parameters: the confidence threshold τ that controls tuple acceptance in Phase 1, the fusion weight α that balances semantic similarity and token overlap in the confidence calibration, and the entropy regularization coefficient λ that weights the semi-supervised loss term in Phase 3.

Each parameter is varied over the range [0.1, 0.9] while holding the other two at their optimal values ($\tau = 0.5$, $\alpha = 0.6$, $\lambda = 0.1$). To account for performance variance caused by random weight initialization, each configuration was independently run three times with different random seeds. The reported accuracy values represent the mean of these three runs. The results are presented in Table 16. For the confidence threshold τ , the accuracy remains stable within 2.5 percentage points over the interval [0.1, 0.9], with a maximum of 0.912 at $\tau = 0.5$. Performance degrades noticeably at extreme values: when $\tau = 0.1$, low-quality tuples are admitted and accuracy drops to 0.878; when $\tau = 0.9$, overly strict filtering removes useful extractions, and accuracy declines to 0.884. For the fusion weight α , the accuracy varies within 1.8 percentage points across [0.3, 0.7], confirming that the hybrid fusion is not sensitive to the precise balance between semantic and token-level similarity, provided that both signals contribute meaningfully. Setting $\alpha = 1.0$ (semantic similarity only) reduces the accuracy to 0.881, while $\alpha = 0.0$ (token overlap only) yields 0.875, indicating that each metric alone is less effective than their combination. For the entropy regularization coefficient λ , the accuracy remains within 2.1 percentage points over [0.4, 0.8], with optimal performance at $\lambda = 0.1$. Excessively large λ values (≥ 0.5) force overconfident predictions on unlabeled nodes and reduce accuracy to 0.867.

TABLE 16. Sensitivity analysis of key parameters.

τ	Accuracy	α	Accuracy	λ	Accuracy
0.1	0.876 ± 0.006	0.0	0.872 ± 0.007	0.00	0.866 ± 0.008
0.3	0.899 ± 0.005	0.2	0.896 ± 0.005	0.05	0.904 ± 0.005
0.5	0.910 ± 0.004	0.4	0.905 ± 0.005	0.10	0.911 ± 0.004
0.7	0.903 ± 0.005	0.6	0.910 ± 0.004	0.20	0.901 ± 0.006
0.9	0.881 ± 0.007	0.8	0.901 ± 0.006	0.50	0.864 ± 0.009

Overall, the urgency accuracy varies by at most 3.7, 3.8, and 4.7 percentage points across the explored ranges of τ , α , and λ , respectively. These modest fluctuations confirm that the ST-CKG framework is not excessively sensitive to its key hyperparameters. Notably, the accuracy at $\lambda = 0.00$ (0.866) is statistically indistinguishable from that at $\lambda = 0.50$ (0.864), despite representing fundamentally different failure modes: the former suffers from insufficient supervision for unlabeled nodes, while the latter overfits to noisy pseudo-labels enforced by extreme regularization. This observation further confirms that the reported results reflect the intrinsic capability of the architecture rather than fortuitous parameter tuning.

In summary, the ablation results confirm that each component of ST-CKG—the LLM soft verifier, spatio-temporal hyperedges, and rule distillation—contribute non-trivially to the overall performance and that the full framework offers a practical balance between accuracy, robustness, and operational cost. The sensitivity analysis further demonstrates that the framework’s performance is stable across a broad range of hyperparameter settings, supporting its suitability for real-world deployment, where extensive per-dataset tuning may not be feasible.

E. CASE STUDY

To qualitatively illustrate the end-to-end reasoning process of the proposed ST-CKG framework and demonstrate its interpretability, we present a detailed walk-through using real citizen complaints from the test set. This case study traces the full pipeline of unstructured text through structured extraction, hypergraph construction, and neurosymbolic inference, and concludes with attention visualization, which reveals the evidential basis of the system’s decisions.

1) CASE DESCRIPTION

Complaint ID “Hotline-210214” was filed on February 14, 2021, at 11:22 a.m. from “North Gate of Ruyuan Community, X Town.” The original complaint text, translated from Chinese and paraphrased for clarity, reads: “There is a brick-and-tile structure built without a permit near the north gate of the community. This structure is currently used for rental accommodation and educational tutoring purposes, and constitutes an unauthorized construction. This affects the appearance of the community and may pose a safety hazard to residents and visitors.” Complaints were categorized under the “Urban Management” thematic type. According to the official resolution record, the case was handled by a Comprehensive Law Enforcement Team in collaboration with the

village and property management committees. The resolution involved an on-site inspection, a special meeting convened to address unauthorized construction within the community, and the issuance of a demolition order.

This case was selected for detailed analysis for two reasons. First, it falls within the rule-uncovered subset: no existing SWRL rule chain in the symbolic teacher’s knowledge base matches its specific combination of entity type (“brick-and-tile structure”), hazard description (“unauthorized construction for dual residential and commercial use”), and spatial context (“community north gate entrance”). Therefore, the Rule-only baseline produced no output for this complaint, whereas the actual resolution record demonstrated that correct handling is possible through manual expert reasoning. Second, the case involves a rich spatiotemporal context: it occurred during winter in a community with recent similar complaints, making it well-suited for demonstrating the contribution of both spatial and temporal hyperedges to the inference process.

2) PHASE 1

Hybrid Information Extraction: The Backbone models (fine-tuned UIE for span extraction and BERT for risk classification) process the complaint text and produce structured fields, as summarized in Table 17. The fine-tuned UIE extracts “brick-and-tile structure” as the entity, “unauthorized construction” as the hazard, “used for rental and education” as the event, and “safety hazards and community appearance impact” as the consequence. The fine-tuned BERT classifies the risk type as “regulation violation risk” with a softmax probability of 0.87.

The LLM soft verifier (LLaMA-3-8B-Instruct) independently processes the same complaint text and produces candidate extraction. The two models agree on all fields with high consistency: semantic similarity scores exceed 0.90 for the entity, hazard, and event fields; the sole minor discrepancy occurs in the hazard field, where the LLM suggests the broader term “illegal construction” while the Backbone extracts the more specific “unauthorized construction.” The fusion algorithm computed a mean confidence score of $c = 0.89$, which is well above the acceptance threshold of $\tau = 0.5$. This high confidence indicates that the extracted tuples are reliable and require no human review before proceeding to the subsequent phases.

3) PHASE 2

Spatio-Temporal Hypergraph Integration: The extracted tuples are mapped onto the Community Governance Ontology (CGO) and integrated into the spatio-temporal causal hypergraph. Beyond the standard hyperedge, which groups the complaint with its entity, hazard, risk, event, and consequence nodes, the hypergraph construction procedure establishes five additional hyperedges that connect the complaint to relevant historical cases.

Three spatial proximity hyperedges were established with prior complaints that satisfied the following three conditions: (a) occurring within the same administrative district

TABLE 17. Extraction and inference results for case study Hotline-210214.

Component	Field Extracted/Inferred	Value
Basic Information	Complaint ID	Hotline-210214.
	Timestamp	2021-02-14 11:22 (Winter, Midday). X Town, Ruyuan Community North Gate.
	Location	Thematic Category Urban Management.
Phase 1 Extraction	Entity	Brick-and-tile structure.
	Hazard	Unauthorized construction.
	Risk (Inferred)	Regulation violation risk.
	Event	Structure used for rental and education.
	Consequence	Safety hazards and community appearance impact.
	Mean Confidence c	0.89
Phase 2 Hyperedge Connections	$E_{standard}$	{Complaint, Entity, Hazard, Risk, Event, Consequence}.
	$E_{spatial}(3edges)$	Complaint-1803 (same district, $\Delta t = 3$ days). Complaint-1921 (same district, $\Delta t = 5$ days). Complaint-2015 (same district, $\Delta t = 7$ days).
	$E_{temporal}(2edges)$	Complaint-1042 (Winter, unauthorized construction). Complaint-1567 (Winter, building safety hazard).
Phase 3 Inference	Urgency	High.
	Responsible Department	Comprehensive Law Enforcement Team.
	Collaboration Network	Village Committee & Property Management Committee.

(M Subdistrict, as determined by the administeredBy property of their respective place nodes); (b) sharing the identical risk type (“regulation violation risk”); and (c) having timestamps within a 7-day window of Hotline-210214. Specifically, Complaint-1803 was filed on February 11, 2021 ($\Delta t = 3$ days) and reported an unauthorized canopy extension at a residential building entrance; Complaint-1921 was filed on February 9, 2021 ($\Delta t = 5$ days) and concerned the illegal enclosure of a public corridor on the second floor of a residential building; and Complaint-2015 was filed on February 7, 2021 ($\Delta t = 7$ days) and described an unpermitted structure being used as a storage shed within a residential compound. All four complaints are geographically concentrated within M Subdistrict and all share the regulation violation risk category. This spatial clustering pattern is a strong indicator of systemic unauthorized construction activity in the district, consistent with the operational knowledge that construction violations often occur in bursts within specific neighborhoods.

Two temporal causal hyperedges were established, with prior complaints involving similar hazard types during the winter. Complaint-1042, filed on January 18, 2021, reported an unauthorized rooftop addition constructed during the winter. Complaint-1567, filed on December 12, 2020, reported an illegal balcony enclosure that posed structural safety

risks during the winter. Although the temporal intervals of 27 and 64 days exceeded the 7-day window used for spatial hyperedges, the temporal hyperedge criteria required only the shared season and shared hazard or event type, both of which were satisfied. The winter-season connection is operationally significant: in municipal governance practice, unauthorized indoor construction and renovation activities exhibit documented winter seasonality, as residents undertake structural modifications during colder months when outdoor construction is less feasible.

4) PHASE 3

Neuro-Symbolic Inference: The HyperGAT student model processes the complaint node through two successive layers of hypergraph attention convolution. The initial feature vector of the node was constructed by concatenating the BERT embedding of the complaint text with sinusoidal geographic encodings of its location coordinates and one-hot temporal encodings of its month (February), hour (11 am), and season (winter). The first convolution layer aggregates information from the incident hyperedges, producing an intermediate representation that encodes the risk profile of the complaint. The second layer integrates information from spatial and temporal hyperedges, enriching the representation with contextual signals from the related historical complaints.

The final representation passed through three task-specific classification heads. The urgency head outputs a softmax distribution of (high: 0.91, medium: 0.07, and low: 0.02), yielding a high-urgency prediction. The department assignment head output the maximum probability for the Comprehensive Law Enforcement Team (0.88). The collaboration network head assigned the highest probability to the Village Committee and Property Management Committee (0.85). All three predictions were consistent with the actual resolution actions recorded for this complaint despite falling outside the coverage of the SWRL rule base.

5) ATTENTION-BASED INTERPRETABILITY

Fig. 3 presents the attention heatmap extracted from the second HyperGAT layer for the focal complaint node. The heatmap rows correspond to the top-10 attended neighboring nodes and hyperedges, sorted by their mean attention weights across the three prediction tasks. The columns correspond to urgency, department assignment, and collaboration network tasks. Darker cells indicate higher normalized attention weights, with the color scale ranging from light (weight close to zero) to dark (weight close to one).

The attention pattern revealed a structured and interpretable decision rationale that differed between the three prediction tasks.

For the urgency task, the three spatial hyperedges collectively received the highest attention weights (0.87, 0.83, and 0.79, respectively), indicating that the model’s high-urgency determination was primarily driven by the recognition of three similar regulation violation complaints concentrated within Subdistrict M during the preceding seven days. This

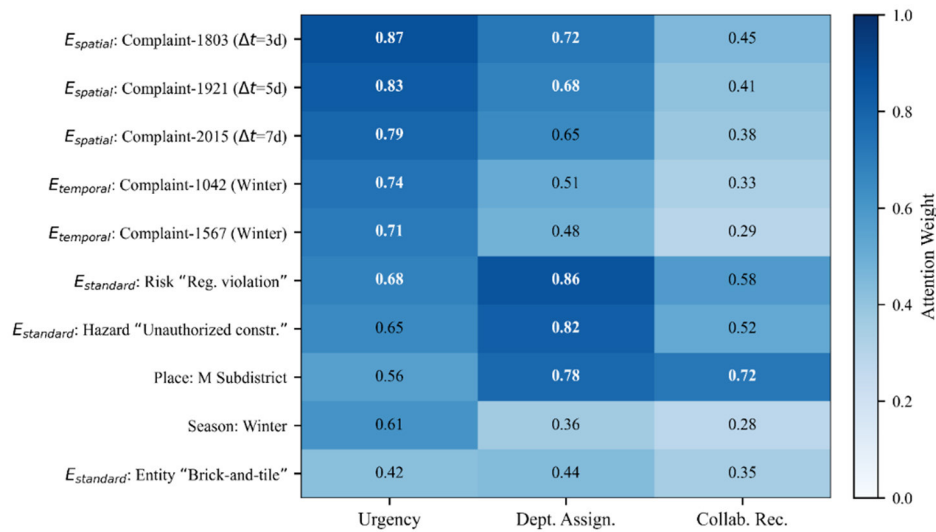


FIGURE 3. Attention heatmap from the second HyperGAT layer for case study Hotline-210214. Rows list the top-10 attended neighboring nodes and hyperedges, sorted by mean attention weight across the three prediction tasks. Columns correspond to urgency, department assignment, and collaboration network recommendation. Darker cells indicate higher normalized attention weights. The spatial hyperedges to prior unauthorized construction complaints in the same district dominate the urgency prediction, while the risk and hazard nodes drive department assignment.

attention distribution aligns closely with the operational logic embedded in the spatial hyperedge construction criteria: a cluster of temporally proximate, geographically concentrated, and risk-homogeneous complaints signals an active, systemic pattern of unauthorized construction that demands prioritized intervention. The temporal hyperedges to winter cases contributed secondary evidence (weights 0.74 and 0.71), which is consistent with the known seasonality of unauthorized indoor construction activities. The Season node itself receives moderate attention (0.61), further confirming that the winter context meaningfully contributes to the urgency elevation.

For the department assignment task, the Risk node ("regulation violation risk") and the Hazard node ("unauthorized construction") receive the two highest attention weights (0.86 and 0.82 respectively). This pattern mirrors the decision logic encoded in the SWRL participant assignment rules that specify that the responsible department is determined by mapping the risk type to the department whose responsibility type covers that risk. The Place node ("M Subdistrict") receives substantial attention (0.78), reflecting the jurisdictional constraint: a department can only be assigned if it has authority over the district where the complaint occurred. The spatial hyperedges received moderate attention (0.65–0.72), suggesting that the model also considered departments that handled similar prior complaints in the same area. The model effectively internalizes the multifactor decision logic through knowledge distillation.

For the collaboration network task, the place node received the most attention (0.78), which is consistent with the SWRL collaboration rules that determine a collaboration network based on administrative jurisdiction. Risk node (0.58) and

hazard node (0.52) contribute secondary weights, reflecting the additional constraint that the collaboration network's supporting responsibility type must be compatible with the primary department's responsibility type.

Thus, attention visualization provides a transparent, auditable rationale for each element of a system's decision. A human operator reviewing this recommendation can interpret the evidence as follows: (i) three similar unauthorized construction complaints were filed in the same district within the past week, warranting heightened urgency; (ii) the Comprehensive Law Enforcement Team is the appropriate responsible department given the regulation violation risk type and its jurisdiction over M Subdistrict; and (iii) the village and property management committees are appropriate collaboration partners because their supporting responsibilities align with the enforcement mandate. This interpretability distinguishes ST-CKG from non-transparent neural systems, and supports the trust and accountability requirements of governmental deployment.

6) IN CONTRAST TO THE RULE-ONLY BASELINE

For Complaint Hotline-210214, the purely symbolic rule engine produces no output: no urgency assessment, department assignment, or collaboration recommendation. Failure occurs because no SWRL rule antecedent successfully matches the combination of a "brick-and-tile structure" entity, an "unauthorized construction for rental and educational use" hazard, and the specific spatial context of a community north gate entrance. Under the original governance workflow, this complaint would remain in an unassigned queue, awaiting manual triage by the human dispatcher. According to the dataset records, the actual

resolution of this complaint, which involved the same departments recommended by the ST-CKG, required approximately 48 hours from the initial filing to the dispatch of an inspection team. ST-CKG produces a matched recommendation in less than one second of end-to-end inference time, preserving the accuracy of expert human judgment while substantially reducing response latency.

In summary, this case study provides qualitative evidence that the ST-CKG framework (i) accurately processes a real citizen complaint through all three phases, (ii) generates correct predictions that match the historical resolution despite falling outside the symbolic rule coverage, (iii) leverages spatio-temporal hyperedges to incorporate relevant contextual information from geographically and temporally proximate cases, and (iv) produces interpretable attention weights that enable human operators to audit and trust the system's recommendations. These properties collectively demonstrate the suitability of the framework for operational deployment in municipal governance.

V. DISCUSSION

The experimental results presented in Section IV provide quantitative evidence of the efficacy of ST-CKG. We now interpret the underlying mechanisms driving performance, assess the current limitations, and outline future research directions.

A. WHY THE HYBRID EXTRACTION APPROACH OUTPERFORMS PURE ALTERNATIVES

The 3–5 percentage point F1 improvement of the hybrid module over the backbone and its larger margin over zero-shot LLMs reflect the principled complementarity. Fine-tuned encoders provide high precision for well-represented entity types, whereas the LLM verifier provides semantic coverage for edge cases involving indirect hazard descriptions and implied consequences that deviate from supervised patterns. Conversely, zero-shot LLMs suffer from hallucinations and schema inconsistencies, grounding their output against the backbone using confidence-aware fusion filters with severe errors, while retaining valid complementary extractions. Thus, the hybrid design realizes a division of labor, where the backbone guarantees schema compliance, the LLM expands the recall of out-of-distribution expressions, and the fusion mechanism arbitrates disagreements by using calibrated confidence scores.

B. WHY NEURO-SYMBOLIC REASONING OUTPERFORMS PURELY SYMBOLIC AND PURELY NEURAL METHODS

In rule-covered cases, the rule-only baseline achieved perfect accuracy, confirming that the SWRL rule base correctly encoded the expert knowledge. In the rule-uncovered cases, 28% of the test sets failed. The neuro-symbolic engine maintains robust performance across both subsets through teacher-student distillation: HyperGAT students learn continuous decision boundaries rather than memorizing discrete rule antecedents, enabling interpolation to structurally similar but

rule-unmatched complaints. The purely neural HyperGAT without distillation overfits the dominant risk categories, with its performance on infrequent types collapsing catastrophically. The symbolic teacher provides an inductive bias that regularizes the learned representation and prevents superficial correlations from dominating decision logic.

C. WHY SPATIO-TEMPORAL HYPEREDGES ARE EFFECTIVE

Removing spatial hyperedges reduces high-urgency recall from 0.88 to 0.72; removing temporal hyperedges reduces it to 0.76. These non-overlapping degradation magnitudes confirm that spatial and temporal signals provide complementary information. Spatial hyperedges enable the detection of emerging risk clusters—multiple similar complaints concentrated in one district within a week—while temporal hyperedges capture seasonal risk recurrence patterns essential for proactive resource allocation. The hypergraph formalism is critical. By encoding spatial and temporal associations as hyperedges connecting arbitrary numbers of nodes, the model captures higher-order interactions that binary relations cannot represent directly.

D. LIMITATIONS

However, this study had several limitations that warrant further investigation. First, hypergraph construction scales with the data volume. At deployment scales of tens of thousands of daily complaints, full-graph training may exceed the GPU memory, requiring subgraph sampling or hierarchical partitioning strategies that are not explored here. Second, the neurosymbolic engine inherits the qualities of the symbolic teachers. Errors or omissions in the SWRL rule base may propagate through distillation and the current framework lacks mechanisms for rule validation or correction. Third, the LLM verifier introduces a residual hallucination risk. Although confidence calibration flags 13.2% of the extractions for review, a small fraction of confident yet incorrect outputs may evade detection. Fourth, the batch-mode processing architecture requires periodic retraining to incorporate new data, thereby introducing a freshness-cost trade-off relevant to live-deployment scenarios. Fifth, while the sensitivity analysis in Section IV-D demonstrates robust performance across a broad hyperparameter range, the optimal values of parameters such as τ and α may vary when the framework is transferred to domains with substantially different linguistic characteristics or complaint taxonomies, thus requiring lightweight recalibration in new deployment contexts. Moreover, the evaluation is confined to a single municipal platform owing to data governance restrictions; the generalizability of the framework to other cities with different administrative structures, languages, and complaint categorization schemes remains to be validated. Securing multi-city datasets is a priority for future research. Sixth, the urgency labels derived from time-to-resolution serve as a practical proxy for operational priority but may not capture nuanced risk gradations in edge cases where resolution is delayed for reasons unrelated to risk severity (e.g., inter-departmental coordination

bottlenecks). Future work could incorporate multidimensional urgency annotations from expert panels to refine this labeling scheme.

E. FUTURE RESEARCH DIRECTIONS

The most immediate extension is to incorporate dynamic graph neural networks, such as temporal graph networks, to enable incremental hypergraph updating as complaints arrive in real time, transforming the framework into a live monitoring and early warning platform that strengthens preemptive governance capabilities. Another promising direction is extending the current rule-based causal chain representation to support counterfactual reasoning within the structural causal model framework. By augmenting SWRL rules with probabilistic causal graphs and estimating intervention effects, the framework could answer “what-if” questions—for instance, predicting how urgency levels would change if a specific department were assigned earlier or if seasonal risk factors were mitigated—thereby providing deeper decision support for urban governance. Beyond the current domain, the core pipeline transfers directly to adjacent smart city applications, including social media opinion monitoring, public safety incident analysis, and environmental complaint processing, which require only domain-specific ontologies and rule bases. Improving the symbolic teacher through active learning, where high-confidence student predictions of rule-uncovered cases are reviewed by experts and codified into new SWRL rules, would progressively expand rule coverage while preserving interpretability. Finally, the cost of the LLM verifier can be reduced through model distillation or adaptive invocation strategies that trigger LLM calls only when backbone confidence falls below a threshold.

In summary, ST-CKG integrates knowledge engineering, spatiotemporal modeling, and neurosymbolic learning to advance urban event management. The identified limitations point to clearly defined research trajectories that will further strengthen the framework’s practical impact and broaden its applicability across the smart city landscape.

VI. CONCLUSION

This study presents ST-CKG, a generalized spatiotemporal risk causal chain knowledge graph framework that integrates hybrid LLM-enhanced extraction, hypergraph-based spatiotemporal modeling, and neurosymbolic reasoning to transform unstructured citizen complaints into structured, interpretable, and actionable governance decisions. Extensive experiments on over 2,000 real-world complaints demonstrated that the framework achieves 91.2% urgency assessment accuracy, maintains robust performance in rule-uncovered cases where purely symbolic baselines fail, and provides interpretable evidence for each recommendation through attention visualization, thereby addressing the critical requirement of algorithmic transparency in governmental applications.

Beyond the quantitative gains of the existing methods, the primary contribution of this study is a replicable paradigm

that bridges the gap between unstructured civic narratives and structured governance intelligence. The modular three-phase design allows individual components to be independently adapted to adjacent smart city domains. Current limitations, including the scalability of full-graph training, dependency on the initial rule-based quality, and residual LLM hallucination risk, define a clear research agenda. Promising extensions include dynamic graph neural networks for real-time incremental learning, active learning strategies for continuous rule-based expansion, and cost-optimized LLM invocation mechanisms. By enabling a systematic understanding of citizen complaints and data-driven decision support that remains accountable for human oversight, the ST-CKG represents a meaningful step toward intelligent, transparent, and proactive urban governance.

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